NEURO MCQ AL HARRIF

{PART 2}



وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ

استعن بالله ولا تعجز

اللَّهمَّ لا سَهلَ إلَّا ما جَعَلتَه سَهلًا، وأنتَ تَجعَلُ الحَزْنَ إذا شِئتَ سَهلًا

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HistoLogy

Histological structure of the cerebral and cerebellar cortex

1- Regarding the grey matter of the cerebellar cortex, it is formed of all of the following EXCEPT: {Summar2025}

a. nerve cells b. myelinated nerve fibers c. neuroglia d. blood vessels

2-The molecular layer contains dendrites of all theses cells except ..... {Mid2025}

a. basket cells b. purkinje cells c. granular cells d. ganglion neurons

3- The molecular layer of the grey matter of the cerebellum contains all of the following EXCEPT: {Summar2025}

a. Dendrites of Purkinje cells. b. Dendrites of the Golgi neurons.

c. Axons of basket cells. d. Dendrite of the granular cells.

4- Betz cells present in which of the following layers ..... {Final2025}

a. Outer pyramidal b. Molecular c. Outer granular d. Inner granular

5- Molecular layer of cerebellum contains {Final2024}

A- axons of purkinje cells B- axons of golgi cells

C- cell bodies of granular cells D- cell bodies of basket cells

6- Dendrites of Purkinje cells in the cerebellum are present in {Mid2024}

a) Purkinje cell layer b) granular layer c) molecular layer d) cerebellar white matter

7-The molecular layer of the cerebellum contains which of the following structures?

a. Granule cell bodies. b. Cerebellar glomeruli. c. Golgi cell bodies

d. Purkinje cell bodies. e. Basket cell bodies

8-The molecular layer is formed of all the following EXCEPT: {Formative2024}

a) Cell bodies of the granular cell layer. b) Dendrites , cell bodies and axons of Purkinje cells.

c) Dendrites of the Golgi neurons. d) Dendrites of the Purkinje cells.

9- The molecular layer is formed of all the following EXCEPT {Final2021}

A- Dendrites of the purkinje cells B- Dendrites of the golgi neurons

C- Dendrites and axons of purkinje cells D- Cell bodies of the granular cell layer

E- Both a& d

10-The granular cell layer is formed of : {Formative2024}

a) The Golgi type II cells. b) The cell bodies, axons and dendrites of basket cells.

c) The dendrites of the Golgi neurons. d) The cell bodies of Purkinje cells.

11- The strucures which present in granular cell layer in the crebellar cortex are {Final2021}

A- The cell bodies of purkinje cells B- Petz cells

C- The cell bodies, axons and dendrites of basket cells D- Glial cells

E- The dendrites of the Golgi neurons

12- Cerebellar cortex is formed of the following EXCEPT {Final2021}

A- Nerve cells B- Neuroglia C- Blood vessls

D- Lymph vessels, E- non-myelinated nerve fibers

13-Glomerular appear in ..... {Final2023}

a) molecular cell layer b) granular cell layer c) purkinje cell layer

14-The cerebellar cortex is formed of all of the following EXCEPT:

A. Nerve cells (neurons) B. Non-myelinated nerve fibers

C. Neuroglia D. Lymphatics

15-The molecular layer of the cerebellar cortex is:

A. The innermost layer B. The middle layer

C. The most superficial (outermost) layer D. Composed entirely of white matter

16-The cerebellar cortex contains all EXCEPT:

A. Purkinje cells B. Granule cells C. Blood vessels D. Olfactory bulbs

17-All of the following are components of the cerebellar cortex EXCEPT:

A. Molecular layer B. White matter C. Granular layer D. Purkinje cell layer

18-The cerebellar cortex contains all EXCEPT:

A. Molecular layer B. Purkinje cell layer C. Granular layer D. Basal ganglia

19-The molecular layer contains which of the following?

A. Dendrites of Purkinje cells B. Dendrites of Golgi neurons

C. Cell bodies, axons, and dendrites of basket and stellate cells D. All of the above

20-Climbing fibers synapse primarily with:

A. Granule cells B. Basket cells C. Dendrites of Purkinje cells D. Stellate cells

21-The molecular layer does NOT contain:

A. Deep cerebellar nuclei B. Basket and stellate cells

C. Purkinje cell dendrites D. Axons of granule cells

22-The dendrites of Purkinje cells extend into which layer?

A. Granular layer B. Molecular layer C. White matter D. Deep cerebellar nuclei

23-The granular layer is located:

A. Outermost (superficial) layer B. Middle layer

C. Deepest layer of the cerebellar cortex D. Part of the white matter

24-The Purkinje cell layer is:

A. The outermost layer of the cerebellar cortex

B. The middle, thin layer of the cerebellar cortex

C. The innermost layer D. Composed entirely of white matter

25-The Purkinje cell layer is characterized by:

A. Multiple rows of Purkinje cell bodies B. A single row of widely spaced Purkinje cell bodies

C. Only dendrites, no cell bodies D. Only axons

26-Purkinje cells are shaped like:

A. Stellate B. Pyriform C. Round D. Fusiform

27-The granular layer contains which of the following cell types?

A. Granule cells B. Golgi type II cells C. Both A and B D. Purkinje cells only

28-Granule cells in the granular layer are characterized by:

A. Large dense nuclei and thin cytoplasm (lymphocyte-like)

B. Vesicular nuclei and large cytoplasm

C. Stellate shape D. Pyramidal shape

29-The axons of Purkinje cells in the granular layer:

A. Terminate within the granular layer B. Pass to the white matter

C. Form cerebellar glomeruli D. Synapse with mossy fibers

30-Golgi type II cells have dendrites that:

A. Remain confined to granular layer

B. Extend throughout all three layers of the cerebellar cortex

C. Only project to molecular layer D. Form Purkinje cell axons

31-Which fibers pass through the granular layer to reach the molecular layer?

A. Climbing fibers B. Mossy fibers C. Parallel fibers D. Purkinje cell axons

32-Cerebellar glomeruli are composed of all EXCEPT:

A. Terminals of afferent mossy fibers (center) B. Axon of Golgi type II cells

C. Terminal dendrite of granule cells D. Purkinje cell soma

33-Pyramidal cell layer present in ……

a. Cerebellar cortex b. Cerebral cortex

34-Purkinje cell layer present in ……

a. Cerebellar cortex b. Cerebral cortex

35-Which of the following cell types forms the granular layer of the cerebellar cortex

A. Dendrites of Golgi B. Cell bodies of basket cell

C. Dendrites of Purkinje D. Axon of Golgi

36-The nerve cells of the cerebral cortex are arranged into:

A. Three layers B. Four layers C. Six layers D. Ten layers

37-Which of the following is NOT a component of the cerebral cortex?

A. Neurons B. Non-myelinated fibers C. Neuroglia D. Purkinje cells

38-The outermost layer of the cerebral cortex is:

A. Molecular layer B. Internal granular layer C. Multiform layer D. White matter

39-The innermost layer of the cerebral cortex is:

A. Molecular layer B. Multiform (polymorphic) layer

C. External granular layer D. Internal pyramidal layer

40-Which of the following fibers are mostly found in the cerebral cortex?

A. Non-myelinated fibers B. Myelinated fibers C. Climbing fibers D. Mossy fibers

41-Betz cells are large pyramidal neurons found in which layer of the cerebral cortex?

A. Molecular layer (Layer I) B. External granular layer (Layer II)

C. Internal pyramidal layer (Layer V) D. Multiform (polymorphic) layer (Layer VI)

42-Which layers of the cerebral cortex are most developed in the motor cortex?

A. Layers I and II B. Layers III and V C. Layers II and IV D. Layers V and VI

43- Which layers of the cerebral cortex are most developed in the sensory cortex?

A. Layers I and III B. Layers II and IV C. Layers III and V D. Layers V and VI

44-Concerning cerebellar glomeruli

A. Present in the molecular layer of cerebellar cortex

B. Contains terminal dendrite of granule cell

45-Concerning climbing fibers

A. Present only in granule cell layer of cerebellar cortex

B. Present in granule cell layer of cerebaral cortex

C. Present in all layers of cerebellar cortex

1. B	2. C	3. D	4. A	5. D
6. C	7. E	8. A	9. D	10.A
11.E	12.D	13.B	14.D	15.C
16.D	17.B	18.D	19.D	20.C
21.A	22.B	23.C	24.B	25.B
26.B	27.C	28.A	29.B	30.B
31.A	32.D	33.B	34.A	35.D
36.C	37.D	38.A	39.B	40.A
41.C	42.B	43.B	44.B	45.C

Histological structure of the eye and its accessory structures

1-Melanocyte is present in the following structures EXCEPT: {Formative2026}

a) Choroid b) Cornea c) Eye lid d) Iris

2- The lining epithelium of the cornea is : {Summar2025}

a. Simple squamous b. Simple columnar c. Simple cuboidal d. Stratified squamous

3- The nervous layer of the eye include: {Summar2025}

a. Cornea b. Sclera c. Retina d. Conjunctiva

4- All of the following share in transparency of cornea except ……. {Final2025}

a. It's formed of pseudo stratified squamous epithelium

b. The fibers arranged in same direction c. Absence of keratinization

5- All of the following are characters of lens except {Final2024}

A- transparent B- biconvex C- covered by double layer of lens fibers

6- Which of the following is found posterior to the eye lashes? {Final2024}

a- Glands of Moll b- Sebaceous glands

c- Modified sweat glands d- All answers are correct

7-Posterior surface of the iris is formed of: {Formative2024}

a) Single layer of pigmented epithelium b) Double layer of pigmented epithelium

c) Simple squamous epithelium d) Inner non pigmented and outer pigmented

8-Which layer forms the majority of the cornea? {Formative2024}

a) Substantia propria b) Descemet's membrane c) Endothelium d) Epithelium

9-Which structure is transparent? {Formative2024}

a) Ciliary body b) Choroid c) Cornea d) Ora serrata

10-The epithelium of the cornea is {Formative2024}

a) Simple epithelium b) Stratified squamous epithelium

c) Pseudostratified epithelium d) Columnar epithelium

11-All of the following is vascular EXCEPT: {Formative2024}

a) Sclera b) Conjunctiva c) Retina d) Cornea

12- Regarding the eye lens one is WRONG ? {Final2021}

A- Is a biconvex B- It is transparent

C- It is attached to the ciliary body by the zonule

D- the posterior pole is formed of simple squamous epithelium

E- lens fibers form the main substance of the lens

13- Layer of rods and cones of the retina contains {Final2021}{Final2022}

A- cell bodies of photoreceptors B- dendrites of photoreceptors

C- axons of photo receptors D- cell bodies of amacrine cells E- both a& b

14-The lining of the cornea {Final2023}

a) Stratified squamous b) Simple squamous

c)Stratified columnar d) Simple columnar

15-Which layer forms most of the cornea?

a. Epithelium b. Descemet's membrane c. Endothelium

d. Bowman's membrane e. Substantia propria

16-What are the receptors for vision?

a. Ganglion cells b. Cones c. Rods d. Rods and cones e. Bipolar cells

17-What type of epithelium is on the surface of the cornea?

a. Simple columnar b. Simple cuboidal c. Stratified squamous d. Transitional

18-Which is NOT part of the retina?

a. Cones b. Rods c. Uvea d. Bipolar cells e. Ganglion cells

19-The wall of the eyeball is composed of:

A. Two layers: fibrous and vascular B. Three layers: fibrous, vascular, and nervous

C. Four layers: fibrous, vascular, nervous, and scleral D. Only fibrous layer

20-The outermost layer of the eyeball is:

A. Retina B. vascular layer C. Fibrous layer D. Ciliary body

21-The inner nervous layer of the eyeball is the:

A. Retina B. Cornea C. Iris D. Sclera

22-The fibrous layer of the eyeball consists of:

A. Sclera and cornea B. Choroid and ciliary body C. Retina D. Iris

23-The vascular layer includes all EXCEPT:

A. Choroid B. Ciliary body C. Iris D. Lens

24-The three layers of the eyeball are arranged from outer to inner as:

A. Retina → Choroid → Sclera B. Sclera → Choroid → Retina

C. Cornea → Iris → Retina D. Sclera → Retina → Choroid

25-The anterior transparent portion of the fibrous layer is the:

A. Sclera B. Cornea C. Choroid D. Retina

26-The iris is part of which layer of the eyeball?

A. Outer fibrous layer B. Middle vascular layer (uveal tract)

C. Inner nervous layer (retina) D. Cornea

27-Descemet's membrane is:

A. A layer of fibroblasts in stroma B. An acellular smooth hyaline membrane

C. A vascular layer of cornea D. Outer epithelium

28-Aqueous humor is:

A. A thick viscous fluid B. Part of the vitreous body

C. A lipid-rich fluid D. A thin watery fluid

29-The corneal epithelium is composed of:

A. Stratified squamous non-keratinized epithelium

B. Stratified squamous keratinized epithelium

C. Simple squamous epithelium D. Columnar epithelium

30-The posterior surface of the iris is lined by:

A. Single layer of non-pigmented epithelium B. Two layers of pigmented epithelium

C. Descemet's membrane D. Corneal stroma

31-The stroma (substantia propria) is composed of:

A. Lamellae of parallel collagenous bundles and flattened fibroblasts

B. Stratified epithelium only

C. Acellular smooth membrane D. Endothelium

32-The sclera is opaque because:

A. Its fibers are regularly arranged B. Its fibers are irregularly arranged

C. It contains photoreceptors D. It contains corneal epithelium

33-The sclera constitutes which part of the outer layer of the eyeball?

A. Anterior 1/6 B. Posterior 5/6 C. Entire eyeball D. Only the cornea

34-Which of the following is a feature of the sclera?

A. Highly vascularized B. Transparent

C. Dense irregular connective tissue D. Contains corneal epithelium

35-The muscles in the iris include:

A. Superior and inferior rectus muscles B. Dilator pupillae and constrictor (sphincter) pupillae

C. Ciliary muscles only D. Extraocular muscles

36-Which of the following contributes to scleral opacity?

A. Great water content B. Low water content

C. Presence of Purkinje cells D. Presence of lens fibers

37-The sclera is perforated posteriorly at the:

A. Optic disc B. Lamina cribrosa C. Fovea centralis D. Limbus

38-The central hole in the iris is called:

A. Optic disc B. Pupil C. Fovea D. Limbus

39-The anterior surface of the iris is covered by:

A. Fibroblasts and melanocytes B. Stratified squamous epithelium

C. Endothelium D. Photoreceptors

40-The color of the iris is determined by:

A. The thickness of the stroma only B. Presence of melanin pigment

C. Blood supply only D. Collagen fiber arrangement

41-The choroid is primarily composed of:

A. Dense collagen fibers B. Highly vascular tissue with loose connective tissue

C. Photoreceptors D. Stratified squamous epithelium

42-The rods and cones represent the:

A. Axons of photoreceptors B. Cell bodies of photoreceptors

C. Modified dendrites of photoreceptor cells D. Modified axons of bipolar cells

43-The vitreous humor is:

A. A thin watery fluid B. Part of the cornea

C. A lipid-rich fluid D. A mass of transparent gel

44-The lens is:

A. . A biconcave, transparent, elastic structure B. A biconvex, transparent, elastic structure

C. Part of the vitreous humor D. Part of the cornea

45-The lens is attached to the ciliary body by:

A. Corneal epithelium B. Zonule fibers (suspensory ligaments) C. Iris stroma D. Retina

46-Which of the following is NOT a function of retinal pigment epithelium?

A. Absorption of excess light B. Nourishment of photoreceptors

C. Phagocytosis of outer segments D. Generation of nerve impulses

47-Apical microvilli of retinal pigment epithelial cells surround:

A. Bipolar cells B. Ganglion cells

C. Inner segments of photoreceptors D. Outer segments of rods and cones

48-Lipofuscin granules in RPE represent:

A. Secretory vesicles B. Glycogen stores

C. Residual bodies after phagocytosis D. Lysosomes with enzymes

49-The outer segment of the rod contains:

A. Mitochondria B. Ribosomes C. Transverse membranous discs D. Nissl granules

50-Rods are mainly responsible for vision in:

A. Bright light B. Color vision C. Daylight vision D. Dim light

51-Which of the following cells is more sensitive to light?

A. Cone cells B. Bipolar cells C. Ganglion cells D. Rod cells

52-The rod photoreceptor consists of:

A. Single cylindrical segment B. Outer thin segment and inner broad segment

C. inner thin segment and Outer broad segment D. Inner and nuclear segments

53-The ducts of Meibomian glands open at the:

A. Conjunctival sac B. Corneal surface C. Lid margin D. Lacrimal puncta

54-The outer surface of the eyelid is covered by:

A. Mucous membrane B. Simple squamous epithelium

C. Skin D. Stratified columnar epithelium

55-The inner surface of the eyelid is lined by:

A. Skin B. Periosteum C. Mucous membrane D. Endothelium

56-The glands of Moll are modified:

A. Sebaceous glands B. Salivary glands C. Sweat glands D. Lacrimal glands

57-The glands of Zeiss are classified as:

A. Sweat glands B. Mucous glands C. Serous glands D. Sebaceous glands

58-The lacrimal gland is classified as a:

A. Simple tubular gland B. Compound tubular gland

C. Simple alveolar gland D. Compound tubulo-alveolar gland

59-The conjunctiva lining the inner surface of the eyelid is called:

A. Bulbar conjunctiva B. Fornix conjunctiva

C. Palpebral conjunctiva D. Corneal conjunctiva

60-The conjunctiva covering the anterior surface of the eyeball is known as:

A. Bulbar conjunctiva B. Fornix conjunctiva

C. Palpebral conjunctiva D. Limbal conjunctiva

61-The epithelium of conjunctiva at the fornix is:

A. Stratified squamous without goblet cells B. Stratified columnar epithelium with goblet cells

C. Simple cuboidal epithelium D. Pseudostratified ciliated epithelium

62-The palpebral conjunctiva is lined by:

A. Simple squamous epithelium B. Stratified columnar epithelium

C. Stratified squamous epithelium D. Pseudostratified epithelium

63-The optic nerve is formed from the axons of which retinal cells?

A. Bipolar cells B. Photoreceptor cells C. Müller cells D. Ganglion cells

1. B	2. D	3. C	4. A	5. C
6. D	7. B	8. A	9. C	10.B
11.D	12.D	13.B	14.A	15.E
16.D	17.C	18.C	19.B	20.C
21.A	22.A	23.D	24.B	25.B
26.B	27.B	28.D	29.A	30.B
31.A	32.B	33.B	34.C	35.B
36.A	37.B	38.B	39.A	40.B
41.B	42.C	43.D	44.B	45.B
46.D	47.D	48.C	49.C	50.D
51.D	52.B	53.C	54.C	55.C
56.C	57.D	58.D	59.C	60.A
61.B	62.C	63.D	64.	65.

Histological structure of sensory organs of hearing, balance, taste and smell

1-The Sensory organ of hearing is called: {Formative2025}

a) Organ of Corti b) Crista ampularis c) Macula utriculi d) Macula sacculi

2- Organ of Corti is a neuroepithelial structure which rests on the: {Summar2025}

a. Tectorial membrane b. Basilar membrane

c. Vestibular membrane d. Otolithic membrane

3-Taste buds are located in: {Summar2025}

a. The ventral aspect of tongue and soft palate

b. The dorsum of the tongue and soft palate

c. The anterior surface of epiglottis and soft palate

d. The ventral aspect of tongue and hard palate

4- Gelatinous membrane of organ of corti …… {Final2025}

a. Tectorial membrane

5- The stereo cilia of hair cells are embedded in which membrane {Final2021}

A- Basilar B- Reissner's C- Tectorial D- tympanic

6- Sensory cells in organ of corti {Final2024}

A- hair cells

7-concerning the Sensory organ of balance its receptor cells are present in {Formative2024}

a) the cochlea b) the vestibule

c) both the vestibule and the semicircular canals d) the semicircular canals

8-The ampula of the semicircular canals contains a neuroepithelial structure called: {Formative2024}

a) organ of corti b) crista ampularis c) Maculae utriculi d) Maculae sacculi

9-which type of tongue papillae contain no taste buds {Formative2024}{Final2023}

a) Circumvallate papillae b) Fungifiorm papillae c) Filiform papillae d) Foliate papillae

10- Which of the following is true about taste buds {Final2024}

a. Contains 2 types of cells b. At both sides of filliform papillae

c. Neuroepithelial cells d. on ventral surface of tongue

11- The outer hair cells in the organ of Corti are arranged in {Formative2022}

a) 1 row b) 2 rows c) 3 rows d) 3-5 rows

12- The type of olfactory epithelium is {Formative2022}{Final2022}

a) Simple columnar b) Stratified columnar

c) Pseudo stratified columnar d) Stratified squamous

13- Regarding the organs of equilibrium : {Final2021}{Formative2024}

A- the utricle is lined with stratified epithelium

B- the saccule is lined with simple columnar epithelium

C- both utricle and saccule are not lined with epithelium

D- both utricle and saccules are lined with simple squamous epithelium

E- both b &d

14- Hair cells in the organ of corti {Final2021}{Formative2024}

A- Maintaining equilibrium

B- Have hair like processes which contain microtubules

C- Have hair like processes which contain microfilaments

D- Their basal surface lies directly on the basilar membrane

E- All of the above

15- Taste buds all are TRUE EXCEPT {Final2021}

A- they open into the surface by taste pores B- formed of three types of cells

C- sustintacular cells can compensate other cell types D- found on the sides of tongue papillae-

E- contains neuroepithelial cells

16-Lining of olfactory mucosa {Final2023}

a) Pseudo stratified columnar b) Stratified columnar

17- Which type of cells is responsible for regeneration of taste buds ? {Final2022} {Formative2022}

A- Gustatory epithelium B- Supporting cells C- Basal cells D- Both A&B

18-The sensory organs of balance are located in:

A. Cochlea only B. Middle ear

C. Vestibule and semicircular canals D. Tympanic membrane

19-The Organ of Corti is located within the:

A. Vestibule B. Semicircular canals C. Cochlea D. Middle ear cavity

20-The utricle and saccule are filled with:

A. Perilymph B. Endolymph C. Blood D. Cerebrospinal fluid

21-The Organ of Corti rests on the:

A. Basilar membrane B. Tectorial membrane

C. Reissner's membrane D. Osseous spiral lamina

22-Which of the following structures overlies the hair cells?

A. Basilar membrane B. Tectorial membrane

C. Reissner's membrane D. Tympanic membrane

23-The sensory receptor cells of the Organ of Corti are the:

A. Supporting cells B. Pillar cells C. Hair cells D. Spiral ganglion cells

24-Phalangeal (supporting) cells function to:

A. Produce endolymph B. Form the tectorial membrane

C. Support hair cells and their nerve endings D. Generate receptor potentials

25-Hensen cells are:

A. Sensory hair cells B. Inner pillar cells

C. Supporting columnar cells outside the Organ of Corti D. Bipolar neurons

26-Outer hair cells are arranged in:

A. One row B. 3 to 5 rows C. 6 to 8 rows D. 10 rows

27-Inner hair cells are arranged in:

A. 3 rows B. 5 rows C. One row D. Two rows

28-The basal ends of hair cells rest on:

A. Basilar membrane directly B. Tectorial membrane

C. Cup-like processes of phalangeal cells D. Pillar cells

29-The apical surface of hair cells contains:

A. Motile cilia B. Microvilli only C. Goblet cells D. Stereocilia

30-Which of the following statements is TRUE about stereocilia?

A. They are motile cilia B. They are non-motile cilia

C. They are part of the tectorial membrane D. They are axons of hair cells

31-The lining of the utricle and saccule sacs is:

A. Stratified squamous epithelium B. Columnar epithelium

C. Simple squamous mesothelium D. Pseudostratified epithelium

32-The macula rests on a:

A. Tectorial membrane B. Basement membrane

C. Basilar membrane D. Reissner's membrane

33-The macula is covered by:

A. Basilar membrane B. Otolithic membrane C. Cupula D. Reissner's membrane

34-The macula contains which two main types of cells?

A. Pillar and Deiters' cells B. Hensen and Claudius cells

C. Hair cells and supporting cells D. Ganglion and bipolar cells

35-The stereocilia of hair cells are embedded in:

A. Basilar membrane B. Tectorial membrane C. Cupula D. Otolithic membrane

36-Each semicircular canal has an expanded end called:

A. Macula B. Ampulla C. Crista D. Vestibule

37-The ampulla contains a sensory structure called:

A. Macula B. Tectorial membrane C. Crista ampullaris D. Organ of Corti

38-The receptor organs for taste are located in:

A. Cochlea B. Retina C. Taste buds D. Semicircular canals

39-Filiform papillae are:

A. Rounded with taste buds B. Large with taste buds

C. Sharp, with no taste buds D. Contain olfactory receptors

40-Fungiform papillae are:

A. Sharp, no taste buds B. Rounded with taste buds

C. Large, circumvallate D. Only in pharynx

41-Circumvallate papillae are:

A. Sharp, no taste buds B. Rounded, few taste buds

C. Large papillae with taste buds D. Found in soft palate only

42-Taste buds contain which type of cells?

A. Sustentacular, basal, olfactory cells B. Rods, cones, ganglion cells

C. Gustatory receptor cells, supporting cells, basal cells

D. Hair cells, supporting cells

1. A	2. B	3. B	4. A	5. C
6. A	7. C	8. B	9. C	10.C
11.D	12.C	13.D	14.C	15.C
16.A	17.C	18.C	19.C	20.B
21.A	22.B	23.C	24.C	25.C
26.B	27.C	28.C	29.D	30.B
31.C	32.B	33.B	34.C	35.D
36.B	37.C	38.C	39.C	40.B
41.C	42.C	43.	44.	45.

Pathology

Neurodegenerative and Demyelinating diseases

1- Multiple sclerosis is characterized by: {Summar2025}{سؤال في المحاضرة}

A- Rapid fatal disease. B-Remission and relapse.

C- Regressive course. D-Steady progressive course.

2-Demyelinating disease means: {Formative2024}{Final2024}{Summar2025}

a) Destruction of axons. b) Destruction of the brain stem.

c) Destruction of myelin. d) Destruction of the cerebellum.

3-In multiple/ disseminated sclerosis; the discrete areas of demylinations are called {Formative2025}

a) Aneurysms. b) Amyloid. c) Plaques. d) Oligodendrocytes.

4- Multiple sclerosis is a: {Formative2022}{Final2025}{ثناء}

a) Congenital disease b) Degenerative disease

c) Demyelinating disease d) Metabolic disease

5-Which of the following diseases characterized by remission and relapse ? {Final2025}

a. Multiple sclerosis b. Parkinson's disease c. Alzheimer's disease

6-Parkinson's Disease is {Formative2024} {سؤال في المحاضرة}

a) Inflammatory disease. b) Immune mediated disease.

c) Degenerative disease. d) Demyelinating disease.

7-A post-mortem brain examination of a patient with Parkinson's disease would most likely reveal neuronal loss in the substantia nigra pars compacta and intracellular inclusions known as: {سؤال في المحاضرة}

a) Neurofibrillary tangles b) Lewy bodies (α -synuclein aggregates)

c) Amyloid plaques d) Intraneuronal intranuclear inclusions

8-The primary pathophysiology of Parkinson's disease is the loss of dopaminergic neurons in which part of the brain? {سؤال في المحاضرة}

a. Hippocampus b. Cerebral cortex c. Substantia nigra d. Cerebellum

9-Which of the following are cardinal motor symptoms of Parkinson's disease? {سؤال في المحاضرة}

a. Memory loss and confusion b. Vision changes and fatigue

c. Resting tremor, bradykinesia, and rigidity d. Muscle weakness and spasticity

10-Multiple sclerosis is a disease characterized by the demyelination of the central nervous system. What cells are primarily attacked in this process? {سؤال في المحاضرة}

a) Schwann cells b) Oligodendrocytes c) Astrocytes d) Microglia

11-The neurofibrillary tangles found in Alzheimer's disease are composed of which protein? {سؤال في المحاضرة}

a) α-synuclein b) Tau protein c) β-amyloid precursor protein d) Huntingtin protein

12- multiple Sclerosis is …… {Final2022}{Final2021}

A- Congenital disease B- Inflammatory disease C- degenerative diseases

D- Demylinated disease E- Metabolic disease

13-Multiple sclerosis is characterized by…….. {Final2023}

a) demyelinating areas in spinal cord b) amyloid plaques c) Lewy bodies of dementia

14- Alzheimer's and parkinsonism are {Final2022}

A- Demyelination B- Autoimmune C- Neurodegenerative diseases D- Inflammatory

15- Which part of basal ganglia Is affected in case of Parkinsonism ? {سؤال في المحاضرة} {Final2021}

A- Striatum B- Substantia nigra C- Subthalamic neuron D- Corpus striatum

16- Lewy bodies are present in {Final2021}

A- Alzheimer disease B- Idiopathic Parkinson disease C- Multiple sclerosis

D- All of the above E- None of the above

17- Lewy bodies present in : {Formative2022}

a) Parkinson,s disease b) Alzheimer disease c) Multiple sclerosis d) Amyloidosis

18- Oligodendrocytes functions is : {سؤال في المحاضرة}

A- Phagocytosis. B-Synthesis of CSF.

C-Production of Dopamine. D- Synthesis of myelin

19- Astrocytes are {سؤال في المحاضرة}

A- Neuron B-Glial cells. C- Dural cells. D- Fibroblasts.

20- Parkinson's disease is characterized by: {سؤال في المحاضرة}

A- Lowy bodies B- Langerhan's cells. C- Tangles. D- Plaques.

21- Huntington's Disease is due to dysfunction in: {سؤال في المحاضرة}

A- Substania negra. B- Cerebellum. C- Cerebral cortex. D- Basal ganglia.

22- A 76-years-iold man suffered from depression dementia (recent memory loss and difficulty to stand from sitting position. CA scanning showed... {سؤال في المحاضرة}

A- Brain edema B- Brain atrophy. C- Brain tumor. D- Loss of substantia negra.

23- The Microscopic examination from a biopsy from this patient showed: {سؤال في المحاضرة}

A-Granuloma. B- Plaques and tangles. C- Necrosis. D- Neoplastic cells.

24- The final diagnosis of this patient is.... {سؤال في المحاضرة}

A- Huntington's disease. B- Multiple sclerosis.

C- Parkinson' disease. D- Alzheimer's disease.

25-Alzheimer's disease is a: {ثناء}

A. Congenital disease B. Inflammatory disease C. Degenerative disease

D. Demyelinating disease E. Metabolic disease

26-Which of the following characterizes Alzheimer's disease?

a. Deposition of amyloid in neurotic plaques & in the wall of cerebral vessels

b. Liquefactive necrosis.

c. Decreased number of dopaminergic neurons in substantia nigra.

d. Atrophy in the caudate and putamen.

27-Which of the following is the most characteristic histological finding of Alzheimer's disease?

a. Neurofibrillary tangles. b. Senile plaques. c. Lewy bodies. d. Hirano bodies.

28-Which of the following mechanisms best explain the pathogenesis of multiple sclerosis?

a. Ionizing radiation. b. Metabolic error. c. Immune reaction d. viral infection.

1. B	2. C	3. C	4. C	5. A
6. C	7. B	8. C	9. C	10.B
11.B	12.D	13.A	14.C	15.B
16.B	17.A	18.D	19.B	20.A
21.D	22.B	23.B	24.D	25.C
26.A	27.B	28.C	29.	30.

TUMOURS OF THE CNS

1-Psammoma bodies are characteristic of: {ثناء}{Summar2025}

A. Astrocytoma B. Oligodendroglioma C. Medulloblastoma

D. Meningioma E. Ependymoma

2-Which of the following tumors characterized by verocay bodies ? {Final2025}

a. Meningioma b. Schwannoma c. Medulloblastoma

3-……..is not a nerve sheeth tumor. {Formative2025}

a) Schwannoma b) Neurilimmoma c) Neurofibroma d) Ependymoma

4-……..is not a neuroglial tumor. {Formative2025}

a) Ganglioneuroblastoma b) Astrocytoma c) Oligodendroglioma. d) Ependymoma

5-……. arises from neuroglial cells. {Formative2025}

a) Ganglioneuroma b) Medulloblastoma. c) Neuroblastoma. d) Astrocytoma

6-The following is incorrect as regards to neurofibroma.. {Formative2025}

a) It may be associated with skin pigmentation. b) It may be sporadic or hereditary.

c) It arises from meningeal cells. d) It contains large amount of collagen.

7-Neurilemmoma is {Formative2025}

a) Schwannoma b) Neuroblastoma. c) Ganglioneuroma d) Neurofibroma

8-The following is incorrect in meningioma. {Formative2025}

a) It may arise from lateral cerebral convexities. b) It shows a slight male predominance.

c) It arises from cap cell layer of arachinoid. d) It is usually well-circumscribed.

9-The most common brain tumors is/are: {Formative2025}

a) schwannomas b) ependymoma c) Metastasis d) gliomas

10- Pilocytic astrocytoma is characterized by all EXCEPT {Final2021}{Summar2025}

A- Usually occurs in children B- Usually occurs in the cerebellum C- It is usually cystic

D- It is of bad prognosis E- Rosenthal fibers

11-Microscopic Homer-Wright rosette formation is a characteristic feature of: {Summar2025}

A. Astrocytoma B. Oligodendroglioma C. Medulloblastoma D. Meningioma

12- All of the following are true about Medulloblastoma Except: {Summar2025}

A. Type of glioma B. Presence of malignant small round cells

C. Embryonic tumor D. WHO grade IV

13-All are gliomas except: {Formative2025}

a) meningiomas b) oligodendroglioma c) Astrocytomas d) ependymoma

14- All of the following are true about medullary blastoma except …… {Final2025}

a. It forms both pseudo rosettes and true rosettes b. It forms true rosettes only

15- Which of the following is low grade astrocytoma caused by bipolar cells and fibrillary hair like process ? {Final2025}

a. Pilocytic astrocytoma b. Glioblastoma multiform c. Anaplastic astrocytoma

16- Atypical meningioma is considered WHO grade ? {Final2021}

A- I B- II C- III D- IV E- None of the above

17- The characteristic feature of Glioblastoma multiforme is {Final2021}

A- Microvascular proliferation B- pseudo palisading necrosis

C- Nuclear Atypia D- All of the above E- None of the above

18- The extradural spinal cord tumors could arise from {Final2021}

A- Meninges B- Bone C- Nerves D- None of the above E- All of the above

19- Microscopic rossete formation is a characteristic feature of : {Final2022}

A- Astrocytoma B- Oligodendroglioma C- Medulloblastoma

D- meningioma E- Schwannoma

20- Antoni type A/B is found in {Final2022}{Final2024}

A- Ependymoma B- Meningioma C- Medulloblastoma D- schwannoma

21- Homer- wright rossete in {Final2022}

A- Ependymoma B- Meningioma C- Medulloblastoma D- Schwannoma

22- The prognosis of primary brain tumours depends on: {Final2024}

A- The site of the tumour B- The grade of the tumour

C- The cell of origin of the tumour D- The age of the patient

23-A 65-years-old male complained of headache and Rt hemiparesis (weakness of the Rt side of his body MRI showed a tumor mass that eroded the skull bone. Excisional biopsy of the tumor revealed spindle cells, arranged in syncytial pattern with whorly appearance and psammoma bodies (calcifications at is the most probable diagnosis? {Formative2024}{Final2024}

a) Astrocytoma. b) Meningioma c) Oligodendroglioma. d) Ependymoma.

24-A male patient 40 years old, presented with hemiplegia and deafness. Radiologic examination shows a tumor at the cerebellopontine angle. Biopsy from this tumor shows a neurogenic tumor formed of hypercellular and hypocellular areas. The most possible diagnosis is: {Final2023}

a) Neuroblastoma b) Schwannoma c) Ganglioneuroma d) Neurofibroma

25-Meningioma shows…….. {Final2023}

a) whorly appearance b) bipolar cells with hair processes

26-Glioblastoma multiform is…… {Final2023}

a) grade 1 b) grade 2 c) grade 3 d) grade 4

27-A 76-years old female presented with headache and vomiting for 4 days. MRI showed multiple small lesions in the Rt temporal lobe of the brain. A biopsy from one of these lesions showed neoplastic epithelial cells arranged in acini. What is the most suggestive diagnosis? {Formative2024}{Final2024}

a) Astrocytoma. b) Glioblastoma multiform.

c) Metastatic carcinoma. d) Ependymoma.

28- A patient suffered from deafness, ataxia and hemiplegia Examination showed a well-circumscribed encapsulated tumor mass attached to the 8th cranial nerve.The tumor is characterized by the following EXCEPT {Final2021}

A- Could be seen with neurofibromatosis type II

B- Alternating hypo cellular with hypercellular areas C- Common turn malignant

D- Good prognosis after resection E- Verocay bodies

29- A child presents with a cerebellar tumor that consists of malignant small round cells with rosette formation The most possible diagnosis is {Final2021}

A- Meningioma B- Gliablastoma multiform C- Diffuse astrocytoma

D- Oligodendroglioma E- Medulloblstoma

30-Microscopic rosette formation is a characteristic feature of: {ثناء}

A. Astrocytoma B. Oligodendroglioma C. Medulloblastoma

D. Meningioma E. Schwannoma

31-A 5 years old child suffeed from headache and blurring of vision. Radiologic examination showed a cerebellar tumor. Microscopic examination of a biopsy specimen from this tumor revealed diffuse infiltration by malignant small round cells with rosette formation. The most possible diagnosis is: {ثناء}

A. Anaplastic astrocytoma. B. Glioblastoma multiforme. C. Ependymoma.

D. Neuroblastoma. E. Medulloblastoma.

32-A 10 years old child suffered from severe headache and vomiting. Radiologic examination showed a firm tumor tissue occupying the fourth ventricle of the brain. Biopsy revealed infiltration by small cells with rosette formation. The most possible diagnosis is: {ثناء}

A. Neuroblastoma. B. Ependymoma. C. Medulloblastoma.

D. Oligodendroglioma. E. Astrocytoma.

33-Astrocytoma is a:

a. Nerve sheath tumour. b. Neuronal tumour. c. Malformative tumour.

d. Tumour of undifferentiated cells. e. Glial tumour

34-A 65-year-old man develops new-onset seizures and headaches. A CT scan reveals a 6-cm left-sided intracerebral mass. Which of the following is the most likely diagnosis?

a. Schwannoma. b. Pilocytic astrocytoma.

c. Glioblastoma multiforme. d. Meningioma.

35-A 10-year-old girl develops ataxia and hydrocephalus. CT scan shows a midline cerebellar mass. Which of the following is the most likely diagnosis?

a. pilocytic astrocytoma b. Astrocytoma c. Medulloblastoma

d. Neurofibroma e. Meningioma

36-A 45-year-old woman has had unilateral headaches on the right for the past 5 months. Physical examination yields no remarkable findings. CT scan of the head revealed mass which is surgically removed and microscopic examination shows elongated cells with pale, oblong nuclei and pink cytoplasm with occasional psammoma bodies. What is the most likely diagnosis?

a. Ependymoma b. Tuberculoma c. Meningioma d. Astrocytoma

37-A 76-year-old man has a single episode of grand mal seizure. On physical examination, he is afebrile and normotensive. Motor strength is intact, and there is no loss of sensation. Cranial nerves are intact. His mental function is not diminished. There is a 1-cm, darkly pigmented skin lesion on the upper back. Brain MRI shows three solid, 1- to 3-cm mass lesions, without ring enhancement or surrounding edema, located at the gray-white junction in the right and left frontal lobes. The cerebral ventricles appear normal in size. What is the most likely diagnosis?

a. Oligodendroglioma. b. Glioblastoma c. Meningioma. d. Metastatic melanoma.

38-A patient with multiple skin nodules and skin hyperpigmentation, biopsy from the nodules shows a benign connective tissue neoplasm composed of spindle cells. The most possible diagnosis is:

a. Schwannoma. b. Meningioma. c. Leiomyoma. d. Neurofibromatosis.

39-Astrocytoma is a: {ثناء}

A. Glial tumor B. Neuronal tumor C. Tumor of undifferentiated cells

D. Nerve sheath tumor E. Malformative tumor

40-Medulloblastoma is a: {ثناء}

A. Glial tumor B. Neuronal tumor C. Tumor of undifferentiated cells

D. Nerve sheath tumor E. Malformative tumor

41-Astrocytoma is NOT: {ثناء}

A. A glial tumor B. The commonest primary brain tumor in adults

C. Always in the cerebral hemispheres

D. Classified into grades according to the degree of nuclear atypia

42-The following histologic picture is NOT characteristic of glioblastoma multiforme: {ثناء}

A. Highly anaplastic tumor cells B. Papillary formations C. Necrosis D. Tumor giant cells

43-Pilocytic astrocytoma is characterised by all EXCEPT: {ثناء}

A. Usually occurs in children B. Usually occurs in the cerebellum

C. It is usually cystic D. It is of bad prognosis

44-Oligodendroglioma is NOT a: {ثناء}

A. Glial tumor B. Cerebral tumor C. High grade tumor D. Well-circumscribed tumor

45-Malignant astrocytoma spreads by: {ثناء}

A. Direct spread B. Blood spread C. Lymphatic spread

D. All of the above E. None of the above

46-The following is NOT a feature of meningioma {ثناء}

A. Usually occurs in adults B. Always a benign tumor

C. Has a whorly cut surface D. Characterised by psammoma bodies

47-The tumor which is formed of two growth patterns (Antoni A and B) is: {ثناء}

A. Astrocytoma B. Oligodendroglioma C. Medulloblastoma

D. Schwannoma E. Ependymoma

48-Gliomas include all EXCEPT: {ثناء}

A. Astrocytoma. B. Glioblastoma multiforme. C. Oligodendroglioma.

D. Medulloblastoma. E. Ependymoma.

49-A male patient 40 years old, presented with hemiplegia and defness. Radiologic examination shows a tumor at the cerebelloperative angle. Biopsy from this tumor shows a neurogenic tumor formed of hypercellular and hypocellular areas. The most possible diagnosis is: {ثناء}

A. Neuroblastoma. B. Schwannoma. C. Ganglioneuroma.

D. Ganglioneuroblatoma. E. Neurofibroma.

50-A skin patient with multiple skin nodules and hyperpigmentation, biopsy from the nodules shows a benign connective tissue neopalsm. The most possible diagnosis is: {ثناء}

A. Lipomatosis. B. Neurofibromatosis. C. Neurofibroma.

D. Schwannoma. E. Leiomyoma.

51-A 70 years old male complaining of headache and vomiting. Radiologic examination revealed multiple space occupying lesions in the temporal lobe of the brain. Biopsy taken from one of these lesions showed sheets of malignat cells which are large with small central nuclei and abundant granular eosinophilic cytoplasm. The most possible diagnosis is: {ثناء}

A. Anaplastic astrocytoma. B. Glioblastoma multiforme.

C. Metastatic bronchogenic carcinoma. D. Metastatic renal cell carcinoma.

E. Metastatic colonic adenocarcinoma.

1. D	2. B	3. D	4. A	5. D
6. C	7. A	8. B	9. C	10.D
11.C	12.A	13.A	14.B	15.A
16.B	17.D	18.E	19.C	20.D
21.C	22.B	23.B	24.B	25.A
26.D	27.C	28.C	29.E	30.C
31.E	32.B	33.E	34.C	35.C
36.C	37.D	38.D	39.A	40.C
41.C	42.B	43.D	44.C	45.A
46.B	47.D	48.D	49.B	50.B
51.D	52.	53.	54.	55.

Physiology

Ascending Reticular Activating System (ARAS) or (RAS) and sleep

1- Concerning sleep: {Summar2025}

a) sleep consists only of non-REM sleep. b) sleep consists only of REM sleep.

c) non-REM sleep is characterized by dreams. d) REM sleep is characterized by teeth grinding.

2-Delta waves can be obtained in EEG of a: {Formative2025}

a) non-REM sleep. b) person who sees a bright light.

c) during REM sleep. d) normal wakeful person.

3- Which of the following occurs in REM phase of sleep {Final2025}

a. Grinding of teeth

4-The ascending reticular formation. All are true, EXCEPT:- {Formative2025}

a) sends impulses to most parts of the cerebral cortex.

b) is activated by collateral branches of sensory neurons transmitting impulses to the thalamus.

c) it relays mainly in specific thalamic nuclei.

d) when stimulated electrically tends to increase alertness.

5-Slow wave sleep is associated with: {Formative2024}

a) dream which can be remembered. b) decrease ABP.

c) increase muscle tone. d) increased respiratory rate.

6-EEG record of relaxed subject, who closes his eyes shows the following wave: {Formative2024}

a) alpha. b) delta. c) theta. d) beta.

7-During REM sleep? {Final2024}

A. muscle tone throughout the body is increased B. Beta waves are observed in EEG.

C. individuals rarely awake spontaneously at its end. D. is a dreamless sleep.

8- During REM sleep ? {Final2024}

A- muscle tone throughout the body is increased. B- Beta waves are observed in EEG.

C- individuals rarely awake spontaneously at its end. D- is a dreamless sleep.

9-concerning EEG of a sleeping person, K complex appear in {Final2023}

a) stage 1 b) stage 2 c) stage 3 d) REM

10- Which of the following is TRUE about REM sleep ? {Final2022}

A- Increase the cerebral blood flow

11- Which of the following is much decreased during REM sleep ? {Final2022}{Formative2022}

A- ABP B- Heart rate C- Muscle tone D- Cerebral blood flow

12-Which of the following EEG waves is observed during deep sleep: {Formative2022}

a) Alpha waves b) Beta waves c) Theta waves d) Delta waves

13- Delta waves in EEG are seen in {Final2021}

A- Deep sleep B- REM sleep C- A wake with eye open D- A wake with eye closed

14-A 43-year-old man sustained a lower back injury that causes severe chronic pain. His physician prescribes benzodiazepine sedation medications to help him sleep. Which response best describes why this man has difficulty sleeping without medication? {Guyton}

A) Depression of the amygdala B) Depression of reticular formation

C) Excitation of the amygdala D) Excitation of reticular formation

15- The ascending reticular formation. All are true, EXCEPT: {د}

a) is activated by collateral branches of sensory neurons transmitting impulses to the thalamus.

b) sends impulses to most parts of the cerebral cortex.

c) when stimulated electrically tends to increase alertness.

d) it relays mainly in specific thalamic nuclei.

16-Delta waves can be obtained in EEG of a: {د}

a-normal wakeful person. b-during REM sleep.

c-person who sees a bright light. d-non-REM sleep.

17- EEG record of relaxed subject, who closes his eyes include the following wave: {د}

a-alpha. b-beta c-theta. d-delta.

18- If the subject think or open his eyes, EEG record wave will be: {د}

a-alpha. b-beta. c-theta. d-Delta

19- All is true about alpha waves of EEG, EXCEPT: {د}

a-disappear when a person solves a problem. b-replaced by beta waves during mental activity.

c-has a frequency of 30 c/sec. d-has an amplitude of 50 uv.

20-Which of the following structures is maximally sensitive to linear movement of head in the horizontal plane? {د}

a. macula of saccule. b. macula of utricle.

c) crista ampullaris of semi-circular canals. d) proprioceptors in the neck.

21-Slow wave sleep is associated with: {د}

a) increased respiratory rate. b) increase muscle tone.

c) decrease ABP. d) dream which can be remembered.

22-During REM sleep: {د}

a) muscle tone throughout the body is increased. b) Beta waves are observed in EEG.

c) individuals rarely awake spontaneously at its end. d) is a dreamless sleep.

23- B waves of the EEG :-

a- are observed during relaxed wakeful state

b- are faster than a waves but slower than theta waves

c-disappear when the person becomes alert

d-are observed during REM sleep

24-Correct statements regarding rapid eye movement (REM) sleep include which of the following?

a. It is characterized by a slow but steady heart rate.

b. It is accompanied by loss of skeletal muscle tone.

c. It is associated with increased in parasympathetic activity.

d. It lasts longer than periods of slow-wave sleep

25-A low- frequency, high amplitude synchronized cortical rhythm occurs during:

a. Narcolepsy. b. Stages 3 & 4 slow wave sleep.

c. Stage 1 & 2 slow- wave sleep. d. REM sleep

26-Which sensory signals are most effective in increasing RAS activity?

a) Visual signals b) Auditory signals c) Olfactory signals d) Gustatory signals

27-How do motor signals from the motor cortex influence RAS activity?

a) By inhibiting voluntary movement b) By helping resist sleep through voluntary movement

c) By increasing pain perception d) By causing muscle relaxation

28-Signals from which lobes related to emotions influence the RAS?

a) Occipital and parietal lobes b) Temporal and frontal lobes

c) Insular and occipital lobes d) Parietal and frontal lobes

29-Extensive damage to the RAS, such as tumors, leads to:

a) Increased alertness b) Decreased RAS activity and possible coma

c) Hyperactivity d) No significant effect

30-Which of the following drugs increase RAS activity?

a) Barbiturates b) Sympathomimetics c) Benzodiazepines d) Opioids

31-Activation of which centers decreases RAS activity and promotes sleep?

a) Motor centers b) Sleep centers c) Sensory centers d) Visual centers

32-The overall effect of increased RAS activity is:

a) Promotion of sleep b) Increased alertness and wakefulness

c) Muscle relaxation d) Decreased sensory perception

33-Which of the following decreases RAS activity?

a) Increased sensory signals b) Reduction of signals from sensory pathways

c) Stimulation of motor cortex d) Use of sympathomimetic drugs

34-General anesthesia causes:

a) Decreased RAS activity b) Increased RAS activity

c) No effect on RAS d) Activation of cerebral cortex

35-Which of the following is NOT associated with decreased RAS activity?

a) Narcotics b) Sleep centers stimulation c) Amphetamine d) General anesthesia

36-Decreased RAS activity results in:

a) Enhanced sensory processing b) Sleep or coma

c) Increased voluntary movement d) Heightened alertness

37-Alpha waves are best recorded from which region?

A. Frontal B. Parieto-occipital C. Temporal D. Parietal

38- Which EEG wave has the highest frequency?

A. Delta B. Theta C. Beta D. Alpha

39- Beta waves are most commonly seen during:

A. Relaxed wakefulness with closed eyes B. Deep sleep

C. Intense mental activity D. Light sleep in children

40- Delta waves normally occur during:

A. awake B. Deep sleep C. emotional stress D. feeding

41- Which EEG wave has the largest amplitude?

A. Alpha B. Beta C. Theta D. Delta

42- Alpha waves disappear when:

A. The person closes their eyes B. The person is drowsy

C. The person is relaxed D. The person becomes alert

43- Theta waves are most commonly recorded from which region?

A. Frontal B. Parietal & temporal C. Occipital D. Central sulcus

44- Which of the following EEG waves is most associated with deep NREM sleep (stage 3 & 4)?

A. Alpha B. Beta C. Delta D. Theta

45- A relaxed adult sitting quietly with closed eyes will most likely show:

A. Beta waves B. Alpha waves C. Theta waves D. Delta waves

46- Increased beta wave activity is most typical in:

A. Drowsiness B. Relaxation C. Mental concentration D. Deep sleep

47- Which wave normally appears during emotional stress in adults?

A. Alpha B. Beta C. Theta D. Delta

48- The lowest frequency EEG waves are:

A. Alpha B. Beta C. Theta D. Delta

49- Theta waves in adults usually indicate:

A. Normal quiet wakefulness B. Concentration and problem solving

C. Deep sleep only D. Emotional stress or early sleep stages

50- If EEG shows high amplitude and very low frequency, this most likely indicates:

A. Beta activity B. REM sleep C. Delta waves D. Awake with eyes closed

51-If the subject think or open his eyes, EEG record wave will be:

A. alpha. B. beta. C. theta. D. delta

52-If you were looking at an EEG recording of a sleeping person and noticed theta waves, which stage of sleep would that indicate?

A. REM B. N1 C. N2 D. N3

53-The appearance of sleep spindles on a sleeper's EEG recording would indicate they are in:

A. REM sleep B. N1 stage sleep C. N2 stage sleep D. N3 stage sleep

54-All of the following about Slow wave sleep (NREM) except

A. Low threshold of awakening B. 80 % of sleep cycle

C. Paradoxical sleep D. Show Parasympathetic activity

55-Delta waves can be obtained in EEG of a:

A. normal wakeful person. B. during REM sleep.

C. person who sees a bright light. D. non-REM sleep.

56-Which of the following is much decreased during REM sleep?

A. ABP B. Heart rate C. Muscle tone D. Cerebral blood flow

57- Non-REM sleep typically occurs:

A. After REM sleep B. At the start of sleep

C. Only in the middle of the night D. Only before waking

58- REM sleep represents approximately what percentage of total sleep time?

A. 10% B. 20% C. 50% D. 80%

59- The usual duration of one Non-REM cycle is:

A. 10 minutes B. 20 minutes C. 60 minutes D. 90 minutes

60- Eye movements during Non-REM sleep are best described as:

A. Rapid eye movements B. Eyes deviate upward

C. No eye movement D. Eyes move horizontally

61- During REM sleep, ABP, HR, RR and metabolic rate are:

A. Decreased B. Unchanged C. Increased D. Absent

62- Which type of sleep shows marked hypotonia of muscles?

A. Stage 1 Non-REM B. Stage 2 Non-REM C. Stage 4 Non-REM D. REM sleep

63- Dreams that are remembered mainly during:

A. Stage 1 Non-REM B. Stage 2 Non-REM C. Stage 4 Non-REM D. REM sleep

64- Sleepwalking and sleep-talking occur during:

A. REM sleep B. Slow-wave Non-REM sleep C. Wakefulness

65- Which of the following is characteristic of Non-REM sleep?

A. High brain activity B. Increased sympathetic activity

C. Decreased ABP and HR D. Muscle atonia

66- REM sleep is sometimes called:

A. Quiet sleep B. Slow-wave sleep C. Paradoxical sleep D. Transitional sleep

67- Which statement about muscle tone in Non-REM sleep is TRUE?

A. Completely absent B. Increased markedly

C. Slightly decreased (hypotonia) D. Alternates between high and low

68- A person who is very hard to wake is most likely in:

A. REM sleep B. Stage 4 Non-REM sleep

C. Stage 1 Non-REM sleep D. Light sleep

69- Which of the following increases during REM sleep?

A. Body temperature regulation B. Muscle tone

C. Parasympathetic dominance D. Brain metabolic activity

70- Sleep spindles and K-complexes occur in:

A. REM sleep B. Stage 1 C. Stage 2 D. Stage 4

71- Beta-like waves on EEG are characteristic of:

A. REM sleep B. Stage 4 sleep C. Stage 3 sleep D. Stage 2 sleep

72-Passive sleep occurs mainly due to:

a) Activation of VLPO b) Fatigue of RAS neurons and loss of sensory stimuli

c) Release of melatonin d) Stimulation of cerebral cortex

73-Active sleep mechanism involves:

a) Fatigue of RAS neurons b) Active inhibition of RAS by subcortical centers

c) Increased sensory input d) Motor cortex activation

74-Which of the following neurotransmitters promotes sleep?

a) Melatonin b) Serotonin c) Histamine d) Both a and b

1. D	2. A	3. A	4. C	5. B
6. A	7. B	8. B	9. B	10.A
11.C	12.D	13.A	14.D	15.D
16.D	17.A	18.B	19.C	20.B
21.C	22.B	23.D	24.B	25.B
26.B	27.B	28.B	29.B	30.B
31.B	32.B	33.B	34.A	35.C
36.B	37.B	38.C	39.C	40.B
41.C,D	42.D	43.B	44.C	45.B
46.C	47.C	48.D	49.D	50.C
51.B	52.B	53.C	54.C	55.D
56.C	57.B	58.B	59.D	60.B
61.C	62.D	63.D	64.B	65.C
66.C	67.C	68.A	69.D	70.C
71.A	72.B	73.B	74.D	75.

Speech

1-In fluent aphasia, the lesion occurs in the: {Summar2025}

a) Primary auditory area b) Occipital area c) Wenicke's area d) Hand skills area

2- Which of the following is the function of categorical hemisphere ? {Final2025}

a. Recall b. Recognition c. Spatial skills d. Telling poes

3- Motor apraxia occurs due to ….. {Final2025}

a. Weakness of the controlling muscles b. Loss of memory of doing skilled movement

c. Lesion in area 4

4-Patient can see object but cannot understand its meaning, the lesion in: {Formative2024}

a) Wernicke's area b) Secondary visual area (visual association area).

c) Primary visual area. d) Exner area.

5- Regarding aphasia one of the following is CORRECT ? {Mid2021}{Final2022}

A- Lesion in Broca's area results in sensory aphasia

B- Lesion in exner's center results in auditory aphasia

C- Lesion in auditory center results in motor aphasia

D- Lesion in visual association area results in visual aphasia

6- From Wernick's area, impulses are transmitted to Broca's, area in which {Mid2021}{Final2023}

A- Vocalizations are produced B- Writing is produced

C- Thinking is produced D- Sleeping is produced

7-An aphasia is most likely to be associated with a lesion of: {Formative2025}

a) The parietal lobe b) The temporal lobe

c) The hippocampus d) The limbic system.

8-Regarding aphasia all of the following is correct except: {Formative2025}

a) Implies impairment of consciousness.

b) Is an impairment of language skills without motor paralysis, loss of hearing or vision.

c) Is called sensory aphasia if the patient does not understand the meaning of the words he hears, sees and uses.

d) Is called motor aphasia if the patient understands what the speech sounds and symbols mean but lacks the higher motor skills needed to express them.

9- Motor aprexia is a lesion in : {Mid2023}

A- Broca's area B- Exener's area

10-The categorical hemisphere is mainly concerned with:

A. Emotional behavior B. Spatial orientation

C. Categorization and symbolization D. Musical ability

11-Lesion of the categorical hemisphere most commonly leads to:

A. Mania B. Language disorders C. Parkinsonism D. Ataxia

12-The dominant (categorical) hemisphere is responsible for:

A. Face recognition B. Language functions

C. Visual-spatial perception D. Emotional expression

13-Which psychological disorder is associated with lesion of the dominant hemisphere?

A. Anxiety B. Schizophrenia C. Phobia D. Depression

14-The categorical hemisphere is commonly located in:

A. Right cerebral hemisphere B. Occipital lobe

C. Left cerebral hemisphere D. Brainstem

15-Symbolization ability is mainly controlled by:

A. Dominant hemisphere B. Limbic system

C. Representational hemisphere D. Cerebellum

16-The representational hemisphere is also known as the:

A. Dominant hemisphere B. Linguistic hemisphere

C. Non-dominant hemisphere D. Motor hemisphere

17- The non-dominant hemisphere is mainly concerned with:

A. categorization and symbolization B. Recognition of faces and objects

C. Language comprehension D. None

18- Lesion of the representational hemisphere causes:

A. Broca's aphasia B. Wernicke's aphasia C. No language disorder D. Global aphasia

19- Emotional state commonly associated with non-dominant hemisphere lesion is:

A. Depression B. Anxiety C. Euphoria D. Apathy

20- Agnosia refers to:

A. Loss of speech B. Loss of memory

C. Inability to recognize stimuli D. Loss of motor power

21- In non-dominant hemisphere lesion, the patient tends to ignore stimuli from the:

A. Ipsilateral side of body B. Contralateral side of body

C. Upper limb only D. Lower limb only

22- Which of the following functions is NOT related to the representational hemisphere?

A. Facial recognition B. Object recognition C. Speech articulation D. None

23- In most right-handed individuals, the representational hemisphere is the:

A. Left hemisphere B. Right hemisphere C. Cerebellum D. Brainstem

24-Fluent aphasia is caused by a lesion in:

A. Broca's area B. Exner's area C. Wernicke's area D. Motor cortex

25-Wernicke's area is located in:

A. Inferior frontal gyrus B. Posterior part of superior temporal gyrus

C. Precentral gyrus D. Angular gyrus

26-Broca's area is located in the:

A. temporal lobe B. frontal lobe C. Precentral gyrus D. Angular gyrus

27-Wernicke's area is mainly responsible for:

A. Motor speech production B. Articulation of words C. Writing of words

D. Comprehension of auditory and visual language

28-Damage to Wernicke's area results in inability to:

A. Speak fluently B. Write letters

C. Understand spoken and written language D. Control tongue movements

29-Visual language comprehension is mainly mediated by:

A. Wernicke's area B. Auditory area C. Broca's area D. Motor cortex

30-Speech in Broca's aphasia is typically:

A. Fluent B. Non-fluent

31-Neologisms are most commonly seen in:

A. Broca's aphasia B. Wernicke's aphasia C. Global aphasia D. Conduction aphasia

32-A lesion of the hand skills area causes:

A. Hemiplegia B. Sensory loss C. Motor apraxia D. Ataxia

33- Motor apraxia is best described as:

A. Loss of muscle power B. Loss of sensation

C. Tremor during movement D. Incoordination of skilled hand movements

34- The primary auditory cortex corresponds to Brodmann areas:

A. 17, 18 B. 22 only C. 41 and 42 D. 4 and 6

35- The main function of the auditory association area is:

A. Detection of sound intensity B. Localization of sound

C. Understanding the meaning of spoken words D. Production of speech

36- Understanding spoken language mainly depends on:

A. Broca's area B. Primary auditory cortex

C. Motor cortex D. Auditory association cortex

1. C	2. A	3. B	4. B	5. D
6. A	7. B	8. A	9. B	10.C
11.B	12.B	13.D	14.C	15.A
16.C	17.B	18.C	19.C	20.C
21.B	22.C	23.B	24.C	25.B
26.B	27.D	28.C	29.A	30.B
31.B	32.C	33.D	34.C	35.C
36.D	37.	38.	39.	40.

The cerebellum

1- Dysdiadochokinesia means {Final2021}{Final2022}{Mid2023}{Formative2025}

A- Inability to perform various components of a complex motor act simultaneously

B- Inability to perform alternate opposite movements successively at a rapid rate

C- oscillatory movement that appears on performing a voluntary movement

D- Failure to perform two synergistic motor acts at the same time

2- About Neocerebellum: {Summar2025}

a) inhibitory to stretch reflex b) responsible for coordination of current occurring movements

c) facilitatory to muscle tone d) connected to vestibular nuclei

3- Adiadochokinesia: {Final2024}

a) is failure of the patient to perform rapid coordinate movements.

b) is due to lesion in caudate nucleus.

c) is due to lesion in subthalamus. d) is due to lesion in red nucleus.

4-Which functional division is responsible for balance ? {Formative2022}

a-Cerebrocerebellum b-Spinocerebellum c-Vestibulocerebellum d-None of the above

5- Loss of the breaking functions of the cerebellum describes {Final2021}{Final2024}

A- Rebound phenomenon B- Drunken gait C- Adidokokinesia D- Dysartheria

6- The manifestations of Neocerebellar syndrome occur on the opposite side of the lesiom ? {Final2021}

A- TRUE B- FALSE

7- The cerebellum ? {Mid2021}

A- Has no role in maintenance of balance

B- Neocerebellum inhibits muscle tone

C- Paleocerebellum facilitates muscle tone

D- Cerebellum has a predictive function during walking

8-Function of archicerebellum {Final2023}

a) Equilibrium b) Skilled movements

9-The cerebellum receives sensory input from the: {Formative2025}

a) eyes b) proprioceptors c) labyrinth d) all of the above

10-The cerebellum is primarily important in controlling:

a. Muscle strength.in a fixed position (tilted or not). b. Muscle coordination.

c. Initiation of motor movement. d. Vibration sense. e. Steriognosis

11- Which part of cerebellum responsible for controlling gross movement ? {Final2025}

a. Archicerebellum b. Paleo cerebellar c. Neo cerebellum d. Folliculonodular lobe

12- All of the following occurs in cerebellar ataxia except …….. {Final2025}

a. Nystagmus b. Rebound phenomenon

c. Positive Romberg test d. Dysdiadochokinesia

13- All are true about cerebellar ataxia except: {Summar2025}

a) Decomposition of movements b) Akinesia c) Negative Romberg sign d) Nystagmus

14- A ballet dancer spins to the left, During the spin, her eyes snap quickly to the left This fast eye movement is {Final2021}

A- nystagmus B- post rotatory nystagmus C- ataxia D- aphasia

15- Regarding causes of Nystagmus one of the following is CORRECT ? {Final2021}{Final2022} {Formative2024}

A- Parkinsonism B- Tabes dorsalis

C- Lesion in visual pathway D- Neocerebeller syndrome

16-Nystagmus occur in…….. {Final2023}

a) neocerebellar lesion b) lesion in visual pathway c) paralysis of ocular muscle

17-In cerebellar lesion we suspect all following expect: {Formative2024}{Final2024}

a) Ataxia b) Hypertonia c) Disturbance of gait d) Nystaumus

18-The inability to perform rapidly alternating movements (dysdiadochokinesia) is associated with lesions of the {د}

A) premotor cortex B) motor cortex C) cerebellum D) substantia nigra E) medulla

19- The functions of the cerebellum include all of the following except: {د}

a) stop the movement at the precise intended point. b) prevents oscillation of movements.

c) planning and timing function. d) Initiation of fine movements.

20- Regarding the cerebellum: {د}

a) has no role in maintenance of balance. b) neocerebellum inhibits muscle tone.

c) poleocerebellum facilitates muscle tone.

d) cerebellum has a predictive function during running.

21- Neocerebellar syndrome in humans is characterized by all, EXCEPT: {د}

a) hypotonia. b) pendular knee jerk c) disturbance of balance. d) static tremors.

22- Adiadochokinesia: {د}

a) is failure of the patient to perform rapid coordinate movements.

b) is due to lesion in caudate nucleus.

c) is due to lesion in subthalamus. d) is due to lesion in red nucleus.

23- The following are neurological features of cerebellar lesion, EXCEPT: {د}

a) abnormal finger-nose test. b) rebound phenomenon.

c) dysdiadochokinesia. d) shuffling gait

24- All about kinetic tremors is true, EXCEPT: {د}

a) occurs during rest. b) is due to neocerebellar lesion.

c) is accompanied with hypotonia. d) disappear during sleep.

25- Neocerebellum syndrome is associated with all, EXCEPT: {د}

a) ataxia b) hypertonia.

c) inability to do complex movements. d) rebound phenomenon.

26-Neocerebellum disease is associated with: {د}

a) shuffling gait. b) drunken gait c) stamping gait d) circumduction gate.

27-A cerebellar lesion is most likely associated with severe loss of equilibrium when the lesion involves: {د}

a) the neocerebellum. b) a small portion of the right hemisphere.

c) a small portion of the left hemisphere. d) the flocculonodular lobe.

28- The following are neurological features of cerebellar lesion, EXCEPT: {د}

a) dysmetria b) lead-pipe rigidity. c) dysdiadochokinesia. d) rebound phenomenon.

29- Concerning the functions of the cerebellum the following statements are true, EXCEPT: {د}

a) the cerebellum co-ordinates the muscle contractions required to execute a movement.

b) the cerebellum directly stimulates motor neurons required to make a movement.

c) the cerebellum receives feedback from muscles that execute the movement.

d) the cerebellum is involved in the planning of a movement.

30-Which of the following structures has a primary function to coordinate rate, range, force, and direction of movement? {د}

A) Primary motor cortex B) Premotor cortex and supplementary motor cortex

C) Prefrontal cortex D) Basal ganglia E) Cerebellum

31-Neocerebellar lesions are associated with:

a. Positive Romberg s sign. b. Hyper-reflexia. c. Clasp-knife rigidity.

d. Hypertonia. e. Staccato speech.

32-The function of paleocerebellum is:

A. Fine motor coordination B. Coordination of movements

C. Facilitatory of the muscle tone D. Planning complex movements

33-The main function of archicerebellum is:

A. Coordination of movements B. Equilibrium

C. Fine motor skill control D. Planning of movement

34-Paleocerebellum is also called:

A. Vestibulo-cerebellum B. Flocculonodular lobe C. Spinocerebellum D. Vermis

35-Neocerebellum is also called:

A. Spinocerebellum B. Flocculonodular lobe C. Vermis D. Cerebrocerebellum

36-The main function of neocerebellum is:

A. Balance and posture B. inhibitory to the muscle tone

C. Planning of movements D. Eye movement coordination

37-Characteristic signs of neocerebellar syndrome include all EXCEPT:

A. Intention tremor B. Dysmetria C. Dysdiadochokinesia D. Flaccid paralysis

38-Sensory ataxia is caused by a lesion in:

A. Cerebellum B. Basal ganglia C. Proprioceptive pathways D. Corticospinal tract

39-Motor (cerebellar) ataxia is caused by lesion in:

A. Sensory cortex B. Neocerebellum C. Spinal cord D. Basal ganglia

40-Which of the following can cause sensory ataxia?

A. Tabes dorsalis B. Pernicious anemia

C. Diabetic peripheral neuropathy D. All of the above

41- Loss of braking function of the cerebellum cause : {Mid2023}

A- Rebound phenomenon B- Drunken gait C- Dysdiodocokinesia D- Dysarthria

1. B	2. C	3. A	4. C	5. A
6. B	7. D	8. A	9. D	10.B
11.B	12.C	13.B	14.A	15.D
16.A	17.B	18.C	19.D	20.D
21.D	22.A	23.D	24.A	25.B
26.B	27.D	28.B	29.B	30.E
31.E	32.B	33.B	34.C	35.D
36.C	37.D	38.C	39.B	40.D
41.A	42.	43.	44.	45.

Basal ganglia

1- Chorea is associated with {Final2021}{Final2024}

A- hypertonia B- hypotonia C- lestion of lenticular nucleus D- All of the Above

2- Which of the following is not a neurotransmitter in basal ganglia? {Final2024}

a- Glutamate b- GABA c- Tyrosine d- Acetyl choline

3- The site of lesion in chorea is {Final2021}

A- Subthalamic nucleus B- Caudate neucleus C- Substantia nigra D- The thalamus

4-Athetosis is associated with

A- hypertonia B- hypotonia C- lestion of Subthalamus D- All of the Above

5-Athetosis is a lesion of

A- Globus pallidus B- Subthalamus C- Caudate & putamen

6- Hemiballismus is a lesion of {Final2021}

A- Red nucleus B- lenticular nucleus C- Substantia nigra D- Subthalamus

7-Hemiballismus is a lesion of: {Final2023}{Formative2024}

a) lenticular nucleus. b) Cudate nuleus. c) Subthalamus d) Substantia nigra.

8- Tremor of Parkinson's disease is due to {Final2021}

A- Hyperactivity of cholingergic neurons B- Hyperactivity of dopaminergic neurons

C- Both D- Neither

9-Disease of the extrapyramidal motor system in Parkinsonism causes: {Formative2025}

a) Muscle paralysis.

b) Tremor which is more obvious when the patient is performing skilled movements.

c) Increased involuntary facial movements during speech.

d) Increased muscle tone throughout the range of passive movement.

10- Motor apraxia occurs due to ….. {Final2025}

a. Weakness of the controlling muscles b. Loss of memory of doing skilled movement

c. Lesion in area 4

11-Chorea is related to……. {Final2023}

a) hypotonia b) lesion in lentiform nucleus c) hypertonia

12- Neurotransmitter in nigrostriatal pathway is {Final2021}{Final2022}

A- GABA B- Glutaminergic C- Cholinergic D- Dopaminergic

13- Symptoms of parkinsonism include all these EXCEPT {Final2021}

A- Intension tremors B- Mask-like face C- Pill-rolling movement D- Shuffling

14- Which of these is not a neurotransmitter in the basal ganglia? {Final2021}

A- GABA ergic fibers B- Glutaminergic fibers C- Cholinergic fibers D- Tyrosine

15- Regarding basal ganglia which of the following is CORRECT ? {Final2021}

A- Is the primary motor area in human B- Its lesion leads to static tremors

C- Its lesion leads to kinetic tremors D- Its lesion leads to drunken gait

16- Concerning causes of akinesia in Parkinsonism which of the following is CORRECT? {Final2021}

A- Lesion in serotonergic neurons B- Lesion in dopamenergic neurons

C- Lesion in cholinergic neurons D- Lesion in adrenergic neurons

17- Degeneration of dopamine-containing neurons in the substantia nigra produces: {د}

a) Choreiform movements. b) hemiballismus.

c) suppression of activity in the subthalamic nucleus. d) parkinsonism.

18- Parkinson's disease: {د}

a) is caused by a lesion in the substantia nigra

b) is caused by lesion in the posteroventral nucleus of the thalamus.

c) is accompanied by kinetic tremors.

d) is accompanied by clasp-knife rigidity.

19- All of the following are features of Parkinsonism, EXCEPT: {د}

a) a tremor at rest. b) a deficiency of dopamine in the substantia nigra.

c) a clasp-knife type of spastic rigidity. d) difficulty in initiating the voluntary movements.

20- Parkinson's disease is characterized by: {د}

a) tremors made worse by movements. b) paralysis of one side of the body.

c) increase of dopamine in the substantia nigra. d) lead-pipe rigidity.

21- Parkinsonism results in all, EXCEPT: {د}

a) akinesia. b) kinetic tremors. c) mask face . d) hypertonia

22- Parkinsonism is associated with: {د}

a) kinetic tremors. b) adidochokinesia. c) rebound phenomenon. d) shuffling gait.

23- Chorea: {د}

a) is due to lesion in the substantia nigra. b) may be due to lesion in the internal capsule.

c) is associated with hypotonia. d) is associated with exaggerated tendon jerk.

24-The portion of the basal ganglia concerned with control of cognitive motor activity (memory guided) involves the following: {د}

a) Putamen. b) caudate nucleus. c) globus pallidus. d) subthalamus.

25- The part of the basal ganglia that release an excitatory transmitter (glutamate) is: {د}

a) Caudate nucleus. b) Putamen. c) Globus pallidus. d) Subthalamic nucleus.

26- parkinsonism results from damage of which of the following: {د}

a) substantial nigra. b) thalamus. c) Subthalamus. d) Cerebellum.

27-…… responsible for subconscious associated movements.

a) Cerebellum b) Basal ganglia c) Cerebral cortex d) Thalamus

28- The caudate circuit is mainly involved in:

A. Execution of movement patterns B. Planning sequencing of movements

C. Reflex motor activity D. Posture control

29- The putamen circuit is specifically responsible for:

A. Planning of sequential motor activity B. Execution of learned motor patterns

C. Emotional behavior D. Oculomotor control

30- The putamen circuit primarily projects back to which cortical area?

A. Prefrontal cortex B. Primary motor cortex C. Visual cortex D. Limbic cortex

31- Input to the caudate nucleus mainly comes from:

A. Primary motor cortex B. Cortical association areas C. Thalamus D. Cerebellum

32- Output from GPi goes next to:

A. Cerebral cortex B. Spinal cord C. Thalamus D. Cerebellum

33- A lesion in the caudate circuit would MOST affect:

A. Initiation of voluntary movement B. Execution of skilled movements

C. Planning complex movement sequences D. Muscle tone regulation

34- A lesion in the putamen circuit would primarily impair:

A. Cognitive planning B. Execution of movement patterns

C. Visual processing D. Auditory reflexes

35- The putamen receives its major cortical input from:

A. Association areas involved in thought B. Visual and auditory cortex

C. Premotor cortical areas D. Limbic structures

36- The basal ganglia circuit that deals with cognitive motor planning is:

A. Putamen circuit B. Limbic circuit C. Caudate circuit D. Oculomotor circuit

37-Damage to the basal ganglia commonly causes which of the following?

a) Loss of tendon reflexes b) Inability to scale movements properly

c) Complete muscle paralysis d) Loss of pain sensation

38-A patient with basal ganglia damage shows crude and poorly coordinated writing. This is due to:

a) Aphasia b) Motor apraxia c) Sensory loss d) Spinal cord injury

39-……. associated with dancing movements

a) Parkinson's b) Chorea c) Athetosis d) Hemibalismus

40-The main neurotransmitter released from the substantia nigra is:

a) GABA b) Glutamate c) Dopamine d) Acetylcholine

41-The globus pallidus primarily releases:

a) Acetylcholine b) Dopamine c) GABA d) Glutamate

42-The subthalamic nucleus uses which neurotransmitter?

a) GABA b) Glutamate c) Dopamine d) Acetylcholine

43-The corpus striatum contains which excitatory neurotransmitter?

a) Serotonin b) Acetylcholine c) Dopamine d) GABA

44-Which of the following is a classic feature of Parkinson's disease?

a) Intention tremor b) Resting tremor c) Flaccid paralysis d) Spasticity

45-Loss of arm swing while walking in Parkinson's disease is due to:

a) Loss of associated movements b) Cerebellar damage

c) Muscle paralysis d) Sensory loss

46-Parkinson's patients typically have which type of rigidity?

a) Flaccid b) Lead-pipe or cogwheel c) Spastic d) Pendular

47-shuffling gait is most commonly associated with

a) Cerebellar ataxia b) Parkinson's disease c) Stroke (hemiplegia) d) Multiple sclerosis

1. B	2. C	3. B	4. A	5. A
6. D	7. C	8. A	9. D	10.B
11.A	12.D	13.A	14.D	15.B
16.B	17.D	18.A	19.C	20.D
21.B	22.D	23.C	24.B	25.D
26.A	27.B	28.B	29.B	30.B
31.B	32.C	33.C	34.B	35.
36.C	37.B	38.B	39.B	40.C
41.C	42.B	43.B	44.B	45.A
46.B	47.B	48.	49.	50.

Memory

1- Procedural memory: {Summar2025}

a) includes skills and habits. b) relates to classical and operant conditioning.

c) facilitation of new words learning. d) processed in the neocortex.

2-Priming is a type of:

a. Episodic memory. b.Short term memory. c.Implicit memory. d. Semantic memory

3-Explicit memory is a type of:

a. Long term memory. b. Episodic memory c. Semantic memory.

d. Short term memory. e. Implicit memory.

4- Priming is the function of which part … {Final2025}

a. Neocortex b. Cerebellum

5- After formation of memory Engram ; we can say that we have {Final2022}{Formative2022}

A- Short term memory B- Declarative memory

C- Long term memory D- Spatial memory

6-Which is TRUE about the declarative memory? {Final2023}

a) Not include consciousness b) Recalling done reflexly

c) Processed in hippocampus

7- Which of the following is responsible for facilitation of recognition of words by prior exposure to term ? {Final2022}

A- Neocortex priming

8- Procedural memories are processed in {Final2022}{Formative2022}

A- Hypothalamus B- hippocampus C- Amyglada D- cerebellum

9-Memory trace formation involves release of which neurotransmitter?

A. Dopamine B. Acetylcholine C. Serotonin D. GABA

10-Short-term memory is primarily mediated by:

A. Long-term potentiation B. Reverberating circuits C. Hippocampus D. Basal ganglia

11-Reverberating circuits are also called:

A. Feedback inhibition B. After-discharge circuits

C. Presynaptic facilitation D. Long-term potentiation

12-Memory of a phone number long enough to dial it is an example of:

A. Long-term memory B. Procedural memory

C. Working memory D. Semantic memory

13-Which of the following enhances Immediate memory temporarily?

A. Presynaptic facilitation B. Reverberating circuits

C. Both A and B D. Long-term potentiation

14-After-discharge in reverberating circuits is associated with:

A. Working memory B. Short term Memory C. Long Term Memory D. Negative Memory

15-Temporal memory traces refer to:

A. Permanent memory storage B. Transient chemical changes in neurons

C. Muscle memory D. Visual memory only

16-Which mechanism contributes to short-term memory?

A. Temporal memory traces B. Permanent adhesion of transmitter to receptors

C. Reverberating circuits D. Axonal myelination

17-Memory engrams are:

A. Temporary neural activity B. Physical and functional changes in synapses

C. Muscular memory D. Short-term facilitation

18-Which of the following is a feature of long-term memory?

A. Temporary Ca^{2+} influx

B. Permanent adhesion of synaptic vesicles to presynaptic membrane

C. Short-term facilitation D. Reverberating circuits

1. A	2. C	3. A	4. A	5. C
6. C	7. A	8. D	9. C	10.B
11.B	12.C	13.C	14.A	15.B
16.A	17.B	18.B	19.	20.

Limbic system

1-The hypothalamus regulates {Formative2024}

a) Muscle tone b) Body equilibrium c) Voluntary movements d) Food intake

2- Which one of the following hypothalamic nuclei is responsible for the detection of the core body temperature? {Final2021}

A- The lateral hypothalamus B- The arcuate nucleus

C- The paraventricular nucleus D- The anterior hypothalamus

3-Fear is caused by stimulation of:

a. Entorhinal cortex and hippocampus. b. Thalamus and hypothalamus.

c. Cerebral cortex and hippocampus. d. Amygdala and hypothalamus.

4- Which of the following is part from limbic system ? {Final2024}

A- thalamus B- amygdaloid nucleus C- meta thalamus

5- The most potent rewerd centre from the following is {Final2022}{Final2023}

A- Medial forebrain

6- Reward and punishment is function of ? {Final2024}

A- limbic system

7- Which part is responsible for Rage? {Final2023}

a. Amygdala

8- The most potent reward center is found in: {Final2025}{Summar2025}

a) The neocortex b) The amygdala

c) The medial forebrain bundle d) The hippocampus

9- Your patient has been diagnosed with an eating disorder. The patient seems to have lost a significant amount of weight in the last year. A lesion in which of the following areas might have caused the patient's problem? {Final2021}{Final2022}{Formative2022}

A- Medial preoptic nucleus B- Paraventricular nucleus

C- Mammillary body D- Ventromedial nucleus of hypothalamus

E- Lateral part of the anterior zone of hypothalamus

10-The thirst center of the hypothalamus contains:

A. Baroreceptors B. Chemoreceptors C. Osmoreceptors D. Glucoreceptors

11-Anterior hypothalamic nuclei primarily control:

A. Sympathetic functions B. Parasympathetic functions

C. Endocrine functions only D. Temperature regulation

12-Posterior and lateral hypothalamic nuclei control:

A. Parasympathetic functions B. Circadian rhythm

C. Sympathetic functions D. Feeding behavior

13-Stimulation of posterior and lateral hypothalamus leads to:

A. Decreased heart rate B. Vasodilation

C. Increased heart rate and arterial pressure D. Hypotension

14-Stimulation of the preoptic area causes:

A. Tachycardia B. Decrease in heart rate and blood pressure

C. Increased sympathetic tone D. Thirst

15-Stimulation of the lateral hypothalamic area causes:

A. Satiety B. Hunger C. Calmness D. Sleep

16-Bilateral lesion of the ventromedial nucleus causes:

A. Loss of appetite B. Obesity due to overactive hunger centers

C. Dehydration D. Hypotension

17-Stimulation of the lateral hypothalamus causes all of the following EXCEPT:

A. Increased eating (hunger) B. Increased thirst

C. Satiety D. Overt rage and increased activity

18-The hypothalamic "master clock" is located in:

A. Lateral hypothalamus B. Ventromedial nucleus

C. Suprachiasmatic nucleus (SCN) D. Posterior hypothalamus

19-Oxytocin secretion is primarily from:

A. Anterior pituitary B. Adrenal medulla

C. Paraventricular nuclei of hypothalamus D. Supraoptic nuclei

20-SCN neurons regulate:

A. Hunger and thirst B. Sleep-wake cycles C. Sexual behavior D. Pain

21-Lateral hypothalamic area lesion leads to:

A. Obesity B. Loss of desire to eat C. Increased thirst D. Hyperthermia

22-Ventromedial nucleus lesion leads to:

A. Loss of appetite B. Dehydration C. Hyperphagia and obesity D. Hypotension

23-Supraoptic nuclei control the secretion of:

A. Oxytocin B. Cortisol C. ADH (vasopressin) D. Prolactin

24-Stimulation of periventricular nuclei leads to:

A. Calmness B. Hunger

C. Fear and punishment reactions (motivation) D. Sleep

1. D	2. D	3. D	4. B	5. A
6. A	7. A	8. C	9. E	10.C
11.B	12.C	13.C	14.B	15.B
16.B	17.C	18.C	19.C	20.B
21.B	22.C	23.C	24.C	25.

Physiology of the eye

1-The circadian rhythm generator in the hypothalamus receives inputs chiefly from: {Formative2025}

a) superior colliculus b) suprachiasmatic nucleus

c) medial geniculate nucleus d) thalamus

2- Which of the following hypothalamic nuclei is responsible for controlling the normal circadian rhythm? {Final2021}{Final2022}{Formative2022}

A- Paraventricular nucleus B- Ventromedial nucleus

C- Suprachiasmatic nucleus D- Lateral nucleus

3-Which of the following is true about Aqueous humor ? {Final2025}

a. It's actively secreted from cells of ciliary processes

4- Myopia cause light rays to fall {Final2024}

a- In front of the retina b- Behind the retina c- On the retina

5- According to Retina Which of the following is true ? {Final2025}

a. In light exposure depolarization of photoreceptors and bipolar cell

b. In light exposure depolarization of photoreceptors only

c. In light exposure depolarization of bipolar cell only

d. Blind spot contains Rods only

6- In dark adaptation {Final2024}

a- sensitivity to light increase b- sensitivity to light decrease d- miosis occur

7- Which of the following statements is TRUE regarding dark adaptation? {Final2022}{Formative2022}

A- It is the process in which eyes get adapted to increased illumination

B- There is reduced sensitivity of rods

C- Resynthesis of rhodopsin occurs. D- The pupil is constricted.

8 - Regarding vision in dim light all of the following are true except {Final2024}

a) rods are more sensitive than cons b) there is decomposition of rhodopsin pigment

9- Regarding Rods & Cons, Rods are: {Final2024}

a- More sensitive to details b- More in fovea centralis

c- More sensitive to colors d- Many Rods synapse with one bipolar cell

10- Regarding Aqueous humor: {Final2024}

a- It is formed by active secretion of Na b- It is formed by active secretion of Cl and HCO3

c- Its drainage is increased during pupil dilatation d- Present in posterior segment of the eye

11- In dark vision {Final2024}

a.Release of neurotransmitters is steady b.RMP is -70

12- Most sensitive area in the eye is….. {Final2024}

a. Fovia centralis

13-The optic disc has no photoreceptors of its own, so is called the …… {Formative2024}

a) A Dead zone b) Blind spot c) Optic nerve d) Lacrimal gland

14-Cells are responsible for capturing light and transforming this into generator potential to be used {Formative2024}

a) the nervous system b) Ganglion cells c) Photoreceptors d) Bipolar cells

15-Photophobia is a light sensitivity when eye open at dim light that stimulates: {Formative2025}

a) Cones b) Rods c) Ganglion cells d) Bipolar cells

16-Rods in the retina are: {Formative2024}{Summar2025}

a) more sensitive to light. b) better innervated than cones.

c) more in the central part of the retina. d) present in fovea.

17-Which of the following is most sensitive to light? {Formative2025}

a) Rods

b) Cones maximally sensitive to light at 565 nm

c) Cones maximally sensitive to light at 535 nm

d) Cones maximally sensitive to light at 440 nm

18-A patient whose eye lens has lost its elasticity is suffering from: {Formative2024}

a) cataract. b) glaucoma. c) astigmatism. d) presbyopia.

19-Color blindness is due to a defect in ……. {Formative2024}

a) Rods and cones b) Cones c) Rods d) Rhodopsin

20- Which of the following statements regarding the Color blindness is FALSE? {Final2022}{Formative2022}

A- It is more common in males. B- It is only inherited

C- It can be inherited or due to acquired conditions

D- It is mostly inherited as a sex-linked recessive character.

21-What are neurons in the retina? {Formative2024}

a) Bipolar b) Pseudounipolar c) Unipolar d) Both a and c

22-The light reflex center is located in the {Formative2024}

a) Pons b) Medulla c) Pretectal region d) Mamillary region

23-The light reflex is integrated in the: {Formative2025}

a) medulla b) frontal eye field c) midbrain d) primary visual area

24- Physiological condition occurs with old age patient can't see near objects … {Final2025}

a. Presbyopia

25- Presbyopia: {Summar2025}

a) the power of accommodation decreases. b) it is caused by large eyeballs.

c) it is caused by small eyeballs. d) reading becomes easier.

26- ……. Is a physiological condition occurring in the eye {Final2022}{Formative2022}

A- cataract B- glaucoma C- presbyopia D- ammetropia

27- Which of the following occurs in light adaptation …… {Final2025}

a. Mydriasis b. Increase sensitivity of rods c. Decrease sensitivity of rods

28-Site of Center of light reflex {Final2023}

a) Midbrain b) Medulla c) Pons d) Diencephalon

29-If light reflex is present and the accommodation reflex absent, then the lesion is most likely in the: {Formative2025}

a) ganglion cells b) pretectal nucleus c) visual cortex d) Edinger-Westphal nucleus

30- All these are TRUE of cones, EXCEPT {Final2021}

A- They are responsible for photopic vision B- They are responsible for colour vision

C- They have rhodopsin D- They are maximal in the macula,

31- In which one of the following sensory systems does stimulation cause the receptor cell to hyperpolarize? {Final2021}{Final2023}{Formative2022}

A- Vision B- Hearing C- Taste D- Touch E- Smell

32- Concerning visual pathway the followings are TRUE EXCEPT {Final2021}

A- It starts in ganglion cells of the retina

B- Optic nerve projects from optic cup at posterior aspect of the eye

C- Optic tract is formed from fibers from ipsi-lateral temporal hemiretina and contra-lateral nasal hemi- retina

D- It ends in the primary visual cortex (Brodmann area 17)

33- Homonymous hemianopia result from the following lesions EXCEPT {Final2021}

A- Optic tract B- Visual cortex C- Optic chiasma D- Optic radiation

34- Concerning the near vision reflex which of the following is CORRECT? {Final2021}

A- Occurs on looking to an object at distance more than 6 meters

B- It is accompanied by mydriasis

C- It is accompanied by decrease in diopteric power of lens

D- It is accompanied by convergence of both eyes

35- Regarding vitreous humor one of the following is CORRECT? {Final2021}

A- It is a watery fluid B- Present in the posterior ocular chamber

C- It is formed and secreted by ciliary process D- It gives support to the lens

36- Concerning myopia which of the following is CORRECT? {Final2021}

A- It occurs usually in small globe

B- due to high refractive power of optical system

C- The light rays focus behind the retina D- Convex lenses correct it

37-Which of the following statements regarding the sclera is FALSE? {Formative2022}

a) It has a protective function. b) It is the tough white fibrous outer layer of eyeball.

c) It gives attachment to the extra ocular ms.

d) It forms the anterior portion of the outer layer of eyeball.

38-The following are functions of aqueous humor, EXCEPT….. {Formative2022}

a) Maintains the shape of eyeball. b) Maintains the intraocular pressure.

c) It holds the retina in its place. d) Provides nutrients, oxygen and electrolytes to the lens.

39-Which of the following cells transmit impulses to the rest of the central nervous system via axons in the optic nerve?. {Formative2025}{Summar2025}

a) ganglion cells b) horizontal cells c) amacrine cells d) bipolar cells

40- The superior colliculus is responsible for {Final2022}{Formative2022}{Final2024} {Summar2025}

A- vision B- hearing C- General touch D- All of the above

41- The photoreceptors resting membrane potential is about …… {Formative2022}

a. 40 mV. b. 70 mV. c. 80 mV. d. 90 mV.

42- Regarding Rods, the following statements are true EXCEPT……. {Formative2022}

a. They are responsible for scotopic vision. b. They have low threshold for light stimulus

c. They do not share in visual acuity. d. They are responsible color vision.

43-Which of the following is photoreceptor? {Formative2022}

a. Rods b. Cones c. Ganglion cells d- both a& b

44- Concerning corneal transparency one of the following is CORRECT {Final2021}{Final2022}

A- It is due to high vascularity of the cornea, B- It is due to rich innervation of the cornea

C- It is due to regular arrangement of corneal layers D- Vitamin D deficiency leads to its loss

45- Regarding ciliary muscles of the eye one of the following is CORRECT {Final2021}

A- It is à voluntary muscle

B- Innervated by parasympathetic nerve fibers

C- Its contraction causes decrease in diopteric power of lens

D- Stimulated by looking to far objects

46- Which of the following is TRUE about visual acuity ? {Final2022}

A- Need stimulation of 2 non-adjacent cones

47- Which of the following statements is CORRECT regarding Hypermetropia? {Final2022}{Formative2022}

A- It is a condition in which parallel rays converge to point in front of the retina.

B- It is corrected by convex lens.

C- It can be due to long eye ball.

D- The vision of near object is not impaired.

48-Which of the following is a step in photoreception in the rods?

A) Light converts all-trans retinal to 11-cis retinal

B) Metarhodopsin II activates transducin

C) Cyclic guanosine monophosphate (cGMP) levels increase

D) Rods depolarize

49-Cutting which structure on the right side causes blindness in the temporal field of the left eye and the nasal field of the right eye? {د}

A) Optic nerve B) Optic chiasm C) Optic tract D) Geniculocalcarine tract

50-Cutting which structure on the left side causes total blindness in the left eye? {د}

A) Optic nerve B) Optic chiasm C) Optic tract D) Geniculocalcarine tract

51-Cutting which structure causes blindness in the temporal fields of the left and right eyes? {د}

A) Optic nerve B) Optic chiasm C) Optic tract D) Geniculocalcarine tract

52-When compared with the cones of the retina, the rods {د}

A) are more sensitive to low-intensity light B) adapt to darkness before the cones

C) are most highly concentrated on the fovea D) are primarily involved in color vision

53- The fovea: {د}

a) is more vascular than the periphery of the retina

b) is composed of cones only

c) contains a greater proportion of ganglion cells

d) contains a greater proportion of rods than the periphery of the retina

54- Aqueous humour: {د}

a) pressure is about 90 mmHg .

b) is easily absorbed when the pupil is widely dilated .

c) formation is dependent on carbonic anhydrase enzyme .

d) is absorbed by the choroids plexus .

55- The primary visual cortex: {د}

a) is located in the temporal lobe.

b) lies adjacent to the calcarine fissure of the occipital cortex.

c) contains areas for form, colour and details of objects .

d) is called V-2.

56- The neural elements responsible for the accommodation of the lens when the eyes are focused on a near object include: {د}

a) the parasympathetic fibers in the trochlear nerve .

b) the parasympathetic fibers in the oculomotor nerve.

c) sympathetic nerve fibers in oculomotor nerve.

d) sympathetic nerve fibers in abducent nerve.

57- The condition in which the lens of the eye becomes totally unaccommodating in elderly persons is: {د}

a) myopia b) emmetropia c) presbyopia d) hypermetropia

58- The aqueous humor is secreted by: {د}

a) iris b) cornea c) lens . d) ciliary processes.

59- Which of the following photoreceptors respond to the longest wavelength of light? {د}

a) rods b) green cones c) blue cones d) red cones

60- Regarding colour vision, which of the following statements is true? {د}

a) the simulation ratio of the three types of cones allows specific colour perception

b) yellow colour is perceived when green and blue cones are stimulated equally

c) when there is no stimulation of red, blue or green cones, there is the sensation of seeing white

d) green is perceived when only green cones are stimulated

61- In response to light, the rods hyperpolarize as a result of: {د}

a) an increase in sodium pump activity of the inner segment.

b) a decrease in Na+ conductance into the outer segment.

c) an increase in potassium conductance into the outer segment.

d) a decrease in sodium pump activity at the inner segment.

62- Lesion of the optic chiasma will lead to: {د}

a) homonymous hemianopia b) bitemporal hemianopia

c) blindness in both eyes d) Protanopia

63- Regarding light adaptation, the following is likely to occur: {د}

a) pupil size increases b) vitamin A converts into retinal

c) rhodopsin concentration is very high

d) photopigments is rods and cones are reduced into opsin and retinal.

64- In myopia: {د}

a) the near point is farther than normal from the eye.

b) near vision is affected more than distance vision.

c) the eye tends to be longer than average from lens to retina.

d) a convex lens can correct the refractive error.

65- Constriction of the pupil increases: {د}

a) refractive power of the eye. b) spherical aberration.

c) depth of focus. d) amount of light entering the eye.

66- During near vision, the following changes occur in the eye, EXCEPT: {د}

a) the visual axes of the two eyes converge. b) The depth of focus increases.

c) The curvature of the cornea increases. d) Chromatic and spherical aberration is decreased.

67- When light is shone into one eye, the pupil: {د}

a) in that eye constricts and in the opposite pupil dilates.

b) constricts even though its optic nerve has been cut

c) responds due to parasympathetic nerve activity

d) responds even after local application of atropine.

68- Presbyopia is caused by : {د}

a) weakness of the ciliary muscle. b) loss of lens elasticity.

c) degeneration of the retina. d) clouding of the cornea.

69- Activation of transducin by light activates an enzyme that: {د}

a) activates adenyl cyclase. b) hydrolyzes cGMP.

c) depolarizes the receptor membrane. d) releases calcium from intracellular stores.

70- The Argyll Roberston pupil: {د}

a) do not constrict because the parasympathetic fibers to the eye are destroyed.

b) do not constrict because the sympathetic fibers to the eye are destroyed.

c) constrict during accommodation when looking at near object.

d) constrict when light is shone on the eye.

71- The fovea of the eye: {د}

a) contains only rods. b) is situated over the exit of the optic nerve.

c) is the region of highest visual acuity. d) is the thickest part of the retina.

72- Which of the following parts of the eye has the highest concentration of rods? {د}

a) fovea. b) peripheral part of retina. c) iris. d) optic disc.

73- Visual accommodation involves: {د}

a) increased intraocular pressure. b) increased tension on the lens ligaments.

c) contraction of the ciliary muscle. d) dilatation of the pupil.

74- In response to light, the following changes occur in the rods, EXCEPT : {د}

a) decreased cGMP b) activation of transduction

c) closure of Na+ channels in the outer segment d) increased release of synaptic transmitter

75- Abnormal colour vision is caused by: {د}

a) recessive gene on the Y chromosome. b) dominant gene on the Y chromosome.

c) recessive gene on the X chromosome. d) recessive gene on chromosome 7.

76- The following event occurs in photoreceptors during phototransduction in response to light: {د}

a) phosphodiesterase activity decreases b) hydrolysis of cGMP increases .

c) neurotransmitter release increases

d) number of open voltage-gated calcium channels increases

77- the electrical events in Rods are started when ……. Is formed: {د}

a) Bathorhodopsin b) Lumirhodopsin c) Metarhodopsin I d) Metarhodopsin II

78- The center of light reflex is: {د}

a) Inferior colliculus b) Pretectal nucleus

c) superior colliculus d) lateral geniculate body

79- Which of the following muscles is contracted as part of the papillary light reflex? {د}

a) dilator pupillae muscle b) constrictor pupillae muscle .

c) ciliary muscle d) medial rectus .

80- in Argyl-Robertson pupil: {د}

a) light reflex is lost and near reflex is intact. b) light reflex is intact and near reflex is lost.

c) Both light and near reflexes are lost. d) Both light and near reflexes are intact.

81- the pupil is constricted during anesthesia in: {د}

a- first stage b) Second stage c) third stage d) fourth stage.

82-Circadian rhythm is the function of:

a. Tuberomammilary nucleus. b. Arcuate nucleus.

c. Suprachiasmatic nucleus d. Lateral nucleus.

83-Cutting the right optic nerve results in?

a. Left homonymous hemianopia. b. Bitemporal hemianopia.

c. Right anopia d. Right homonymous hemianopia.

84-A lesion at the optic chiasm causes?

a. Right homonymous hemianopia. b. Left homonymous hemianopia.

c. Left anopia. d. Right anopia. e. Bitemporal hemianopia

85-A lesion at the left optic tract results in

a. Bitemporal hemianopia. b. Right anopia. c. Left anopia

d. Left homonymous hemianopia. e. Right homonymous hemianopia

86-The space between the iris and ciliary body and the lens is the

a. Posterior compartment. b. Anterior compartment.

c. Posterior chamber. d. The vitreous humor

87-The function of the cornea:

a. Maintains the shape of the eye ball and protects the eye.

b. Acts as the most refractive medium of the eye.

c. Helps in the drainage of the aqueous.

d. Prevents light entry to the eye

88-Regarding the eye:

a- The intraocular pressure is about 1.5 mmHg

b- When the light is shown in one eye, both pupils constrict

c- The pupils dilate when the eye is focused on a near object

d- The aqueous humor is an ultrafiltrate of plasma

89-Glaucoma:

a. Is a disease which causes opacity of lens b. Is a disease which affects far vision

c. Is a disease which produces ischemia of lens

d. Results from increased resistance to fluid outflow through spaces of Fontana

90-The circular muscle of the iris:

a- Contracts in response to bright light to produce pupillary constriction

b- Is innervated by the parasympathetic nervous system

c- Contracts to lax the suspensory ligaments and increase the power of the lens during accommodation

d- Both (a) and (b)

91-One of the following occurs with constriction of the pupil:

a- Increased production of aqueous humor b- Sympathetic nerve fibers are stimulated

c- There is increased depth of focus d- Vision in conditions of poor illumination is improved

e- Chromatic aberration is increased

92-All about meiosis of the pupil is correct, except:

a. Reduces visual field b. Increases depth of field c. Increases the visual acuity

d. Reduces spherical aberration e. Increases chromatic aberration

93-In the corneal reflex:

a- The afferent fibers run in the optic nerve

b- The center lies in the superior colliculus

c- Touching the cornea with a piece of cotton causes pupillary constriction of both eyes

d- Touching the left cornea with a piece of cotton causes blinking of both eyes

94-In order for accommodation to occur:

a- The shape of the cornea must be changed b- The shape of the lens must be changed

c- The ciliary muscle is relaxed for near vision d- The ciliary muscle is contracted for far vision

95-An increase in refractive power is produced by contraction of the:

a. Iris b. Ciliary muscle c. Suspensory ligaments

d. Extraocular muscles e. Pupil

96- The refractive power of the cornea is

a. 14 diopters. b. 18 diopters. C. 42 diopters. d. 60 diopters.

97- dark adaptation includes:

a-The pupil constriction.

b-No Changes in the level of neural activity of the retina.

c-Increase the concentration of photochemical pigments in the retina.

d-These changes require few second to occur

98-Total loss of accommodation in persons over 70 years of age is called:

a- Amblyopia b- Emmetropia c- Hypermetropia

d- Myopia e- Presbyopia

99-Eye in which parallel rays from far object focus behind the retina with accommodation at rest is called

a. Emmetropic eye b. Presbyopic eye c. Myopic eye d. Hypermetropic eye

100-Regarding astigmatism:

a- Light rays emitted from different planes do not come to a common focal point

b- Light rays emitted from distant objects are focused behind the retina

c- Light rays emitted from distant objects are focused in front of the retina

d- Light rays emitted from distant objects are in sharp focus on the retina

e- There is an opaque area in the lens

101-Concerning the pupillary light reflex:

a- The center lies in the superior colliculus

b- Directing the light to one eye causes papillary constriction in this eye only

c- The reflex is absent if the pretectal nucleus is damaged

d- The pathway of this reflex terminates in the visual cortex

102-Concerning the eye all the following is correct, except:

a- The fovea lies on the visual axis

b- Aqueous humour is secreted into the anterior chamber

c- Light reflexly activates the parasympathetic nerves supplying smooth muscle in iris

d- Most of the focusing power is due to the cornea

103-In what layer of the retina is a large store of vitamin A found?

a. The choroid b. The inner nuclear layer.

c. The pigmented layer d. The outer plexiform layer

104-The fovea contains an increased proportion of:

a- Cones b- Ganglion cells c- Horizontal cells

d- Rods e- Vasculature

105-Rods:

a. Are more sensitive to red light b. Are mainly concentrated at the fovea

c. Are primarily involved in color vision d. Have longer receptor potential than cones

106-The fovea of the eye

a- Contains only rods b- Is situated over the exit of the optic nerve

c- Is the region of highest visual acuity d- is the thickest part of the retina

107-Cones:

a. are most abundant in the periphery of the retina. b. are more numerous than rods.

c. have higher sensitivity to light than rods. d. are responsible for color vision.

e. have longer receptor potential than rods

108-When rods are stimulated, the following occurs:

a. Light is necessary for the conversion of all-trans retinal to 11-cis retinal

b. cGMP levels increase c. Rods depolarize

d. Metarhodopsin II activates G protein

109-Regarding retina:

a- Total number of cones in retina is much greater than total number of rods

b- Each cone responds to all wave lengths of light

c- Photoreceptor activation results in hyperpolarization of the receptor

d- Pigment layer of retina contains the photoreceptors

110-After stimulation of photoreceptors, which of the following occurs?

a- Phosphodiesterase activity decreases b- Transducin activity decreases

c- Hydrolysis of cGMP increases d- Neurotransmitter release increases

111-Concerning the retinal photopigments:

a. They are of 3 types

b. They are regenerated in the dark

c. Light is necessary for the conversion of all-trans retinal to 11-cis retinal

d- The photopigment rhodopsin is highly sensitive to red light

112-Which of the reactions in the retinal rods is caused directly by the absorption of light energy?

A- Dissociation of scotopsin and metarhodopsin

b- Dissociation of scotopsin

C- Transformation of 11- cis retinal to all - trans retinal

d- Transformation of metarhodopsin to lumirhodopsin

113-Stimulation of photoreceptors by light produces which of the following:

a. Phosphodiesterase activity decreases b. Formation of metarhodopsin II increases

c. Neurotransmitter release increases d. Activation of rhodopsin kinase enzyme occurs

114-Which of the following pairs of molecules combine to form rhodopsin?

a- Bathorhodopsin and 11-cis retinal b- Bathorhodopsin and all-trans retinal

c- Bathorhodopsin and scotopsin d- Scotopsin and 11-cis retinal

115-Vitamin A deficiency causes:

a- Poor night vision b- Poor color vision c- Astigmatism

d- Presbyopia e- Myopia

116-Cutting which structure on the right side causes blindness in the temporal field of the left eye and the nasal field of the right eye?

a- Optic nerve b- Optic chiasma c- Optic tract d- Primary visual cortex

117-Which of the following conditions is caused by a lesion of the optic chiasma?

a- Bitemporal hemianopia b- Blindness of both eyes

c- Homonymous hemianopia d- Heteronymous hemianopia

118-Which of the following is correct?

a- In Argyl Robertson pupil accommodation reflex is lost and pupillary reflex is intact

b- Astigmatism occurs because the lens elasticity is decreased.

c- Visual aphasia Occurs due to lesion in the cortical areas 18 and 19.

d- Hypermetropia is corrected by biconcave lenses

119- Roberston Argyle pupil disease causes which of the following {Final2025}

a. Loss of accommodation but not light reflex

b. Loss of light but not accommodation reflex

c. Loss of both

d. Both are present

1. B	2. C	3. A	4. A	5. C
6. A	7. C	8. B	9. D	10.A
11.A	12.A	13.B	14.C	15.B
16.A	17.A	18.D	19.B	20.B
21.A	22.C	23.C	24.A	25.A
26.C	27.C	28.A	29.C	30.C
31.A	32.A	33.C	34.D	35.D
36.B	37.D	38.C	39.A	40.A
41.A	42.D	43.D	44.C	45.B
46.A	47.B	48.B	49.C	50.A
51.B	52.A	53.B	54.C	55.B
56.B	57.C	58.D	59.D	60.A
61.B	62.B	63.D	64.C	65.C
66.C	67.C	68.B	69.B	70.C
71.C	72.B	73.C	74.D	75.C
76.B	77.D	78.B	79.B	80.A
81.C	82.C	83.C	84.E	85.E
86.C	87.B	88.B	89.D	90.D
91.C	92.E	93.D	94.B	95.B
96.C	97.C	98.E	99.D	100. A
101. C	102. B	103. C	104. A	105. D
106. C	107. D	108. D	109. C	110. C
111. B	112. C	113. B	114. D	115. A
116. C	117. A	118. C	119. B	120.

Physiology of HEARING

1-The Sensory organ of hearing is called: {Formative2025}

a) Organ of Corti b) Crista ampularis c) Macula utriculi d) Macula sacculi

2- ……. is function of outer hair cells {Final2025}

a. Filtration of the sound

3- Which of the following statements best describes the basilar membrane of the organ of Corti? {Final2022}{Final2023}

A- The apex responds better to low frequencies than the base does

B- The base is wider than the apex

C- The base is more compliant than the apex

D- High frequencies produce maximal displacement of the basilar membrane near the helicotrema

E- The apex is relatively stiff compared to the base

4- The attenuated reflex: {Final2024}

a- Its afferent is trigeminal nerve and efferent is facial nerve

b- Reflex contraction of stapes and tensor tympani in response to loud sounds

c- Is very rapid as its latency is 10 milliseconds

d- Contraction of stapes and tensor tympani to amplify the sound

5- Attenuation Reflex is the major action of: {Summar2025}

a) tensor tympani b) tensor palate c) stapedius d) masseter

6- Your elderly patient has presbycusis that is more evident on the right than the left What might you correctly conclude? {Final2021}

A- The patient has hair cell degeneration at the base of the cochlea

B- The patient has excessive wax buildup in the external auditory meatus

C- The patient has hyperacusis

D- Bone conduction will be better than air conduction on the left,

7-Stimulation of hair cells in the ear is caused by : {Final2023}

a) K influx b) Na influx c) K outflux d) Ca influx

8- The primary function of the bones of the middle ear is to {Final2022}{Formative2024}

A- Amplify the sound stimulus

B- Filter high-frequency sounds from the sound stimulus

C- Enable the direction of a sound stimulus to be detected

D- Enhance the ability to distinguish different sound frequencies

9- Which of the following is the analyzer of hearing ? {Final2022}{Final2025}

A- Basilar membrane

10-Which of the following statements best describes the basilar membrane of the organ of Corti? {د}

A) The apex responds better to low frequencies than the base does

B) The base is wider than the apex

C) The base is more compliant than the apex

D) High frequencies produce maximal displacement of the basilar membrane near the helicotrema

E) The apex is relatively stiff compared to the base

11- The following are functions of the middle ear, EXCEPT: {د}

a) the Eustachian tube equalizes pressure on both sides of the oval window .

b) impedance matching to transfer effectively sounds from the air filled middle ear to the fluid filled cochlea

c) the constant contraction of the middle ear muscles, keeps the tympanic membrane optimally tense for vibration by sounds .

d) attenuation of sound by the reflex contraction of middle ear muscles .

12- Which of the following is the middle ear ossicles that is attached to the tympanic membrane? {د}

a) malleus b) incus c) stapes d) modiolus

13- Tympanic reflex: {د}

a) protects the inner ear from the vibrations of high-frequency sounds

b) increases the rigidity of the ossicular system, thereby reducing conduction of high intensity sounds

c) occurs following a latent period of 4 to 8 seconds after a loud sound

d) can increase the intensity of low-frequency sound transmission by 30 to 40 decibels

14- Regarding the basilar membrane, the following is true: {د}

a) it separates the scala media from the scala vestibule.

b) its fibers decrease in length from the base of the cochlea to the apex of the cochlea.

c) its fibers increase in diameter from the base to the apex of the cochlea.

d) it vibrates best at high frequency near the base of the cochlea whereas it vibrate at low frequency at the apex of the cochlea.

15- The organ of Corti is contained within: {د}

a) saccule. b) scala media . c) scala tympani. d) scala vestibule .

16- Concerning the discrimitation of sound loudness, the following is true: {د}

a) amplitude of vibration of the basilar membrane increases with louder sounds

b) with loud sounds hair cells excite nerve endings at a diminished rate .

c) few outer hair cells are stimulated with loud sounds .

d) few inner hair cells are stimulated with loud sounds .

17- In the middle ear: {د}

a) air pressure is normally subatomspheric.

b) destruction of auditory ossicles abolishes hearing.

c) the round window moves reciprocally with the oval window.

d) paralysis of the middle ear muscles abolishes hearing.

18- The tympanic membrane: {د}

a) stops vibration almost immediately after the sound stops.

b) transmits sound more effectively when the muscles of the middle ear contract.

c) bulges outwards when the Eustachean tube is blocked.

d) modifies the frequency of sound waves impinging on the ear.

19- The primary function of the middle ear bones is to: {د}

a) protect the ear from loud sounds. b) localize sounds

c) amplify sounds. d) frequency discrimination.

20- Depolarization of the hair cells in the cochlea is caused by the influx of: {د}

a) Na+ into the hair cells. b) K+ into the hair cells.

c) Ca++ into the hair cells. d) Cl- into the hair cells.

21- If the patient is unable to hear low-frequency sounds, the damage to the basilar membrane is closest to: {د}

a) oval window. b) helicotrema. c) modiolus. d) spiral ganglion

22- The stereocilia of hair cells are embedded in the: {د}

a) Reissner's membrane: b) basilar membrane .

c) tectorial membrane . d) rods of Corti.

1. A	2. A	3. A	4. B	5. C
6. A	7. A	8. A	9. A	10.A
11.A	12.A	13.B	14.D	15.B
16.B	17.C	18.A	19.C	20.B
21.B	22.C	23.	24.	25.

The vestibular system

1-Which of the following would produce maximum excitation of the hair cells in the macula of both saccules? {Formative2025}

a) Hyperpolarization of the hair cells b) Rotating the head to the right

c) Rapid ascent in an elevator d) Bending the stereocilia away from the kinocilia

2- Saccule is responsible for {Final2025}

a. Angular acceleration b. Linear acceleration in vertical direction

c. Linear acceleration in horizontal direction

3- Which of the following is multidirectional? {Final2024}

A- organ of corti B- utricle and saccule C- crista ampularis D- Pacinian corpuscle

4- Rotation of the head to the right in horizontal canal causes: {Final2024}

a- Hair cells of lateral SCC in right ear are depolarized

b- Hair cells of vertical SCC in left ear are depolarized

c- Hair cells of lateral SCC in right ear are hyperpolarized

d- Hair cells of lateral SCC in left ear are depolarized

5- Which of the following would produce maximum excitation of the hair cells in the right horizontal semicircular canal? {Final2021}{Final2022}

A- Hyperpolarization of the hair cells B- Bending the stereocilia away from the kinocilla

C- Rapid ascent in an elevator D- Rotating the head to the right

6- Which of the following receptors is a unidirectional receptor? {Formative2022}{Final2021}

a. Macula of the saccule and utricle b. Organ of Corti

c. crista ampularis d. Pacinian corpuscle

7- An MD/PhD candidate was doing research on the generation of changes in the membrane potential of cochlear inner hair cells What steps are involved in this process? {Final2021}{Formative2024}

A- When the shorter stereocilia are pushed toward the taller ones, the channel open time is increased, and K+ and Ca2+ enter via the channel and induce hyperpolarization

B- When the shorter stereocilia are pushed toward the taller ones, the channel open time is increased and K+ and Ca2+ enter via the channel and induce depolarization

C- When the taller stereocilia are pushed toward the shorter ones, the channel open time is increased, and Na+ and Ca2+ exit via the channel and induce hyperpolarization

D- When the stereocilia move toward the sound source, the channel open time is decreased, and Na+ and Cl- exit via the channel and induce depolarization

8- The otolith organs (utricle and saccule) are responsible for {Final2021}{Formative2024}

A- Producing the accommodation reflex B- producing vestibuloocular reflex

C- Detecting angular acceleration D- Producing the stretch reflex

9-Which movement lead to maximum stimulation of both saccules? {Final2023}

a) Rapid elevator ascend b) Rotation of head to lateral side

10-Linear acceleration in the vertical plane is signaled by hair cells in: {Formative2024}

a) utricle b) Posterior semicircular canal c) Saccule d) Anterior semicircular canal

11-Linear acceleration in the horizontal plane is signaled by hair cells in:

a) utricle b) Posterior semicircular canal c) Saccule d) Anterior semicircular canal

12-which of these are concerned with rotational acceleration of the head: {Formative2024}

a) Cristae ampullaris b) Maculae utriculi c) Maculae sacculi d) Organ of Corti

13-Depolarization of the hair cells in the cochlea is caused primarily by {Formative2024}

a) flow of K+ into the hair cell b) flow of Cl- out of the hair cell

c) flow of Na+ into the hair cell

14-Which of the following would produce maximum excitation of the hair cells in the right horizontal semicircular canal? {د}

A) Hyperpolarization of the hair cells B) Bending the stereocilia away from the kinocilia

C) Rapid ascent in an elevator D) Rotating the head to the right

15-A ballet dancer spins to the left. During the spin, her eyes snap quickly to the left. This fast eye movement is {د}

A) nystagmus B) postrotatory nystagmus C) ataxia D) aphasia

16- Maintenance of equilibrium depends on information reaching C.N.S. from the following structures, EXCEPT: {د}

a) proprioceptors in muscle and joints. b) cochlear nerve.

c) retina d) vestibular apparatus.

17- The otolith organs are receptors for: {د}

a) linear acceleration b) Angular acceleration. c) proprioception. d) vibration.

18- All are true about the vestibular apparatus, EXCEPT: {د}

a) bony labyrinth contains 3 SCCs. b) helps to maintain equilibrium.

c) is stimulated only during linear acceleration.

d) is stimulated during linear and angular acceleration

19- During rotation in one plane, all are true, EXCEPT: {د}

a) the hair cells of the cristae are stimulated.

b) cristae of the horizontal canals are stimulated when it bends towards the utricle. cristae of vertical canals is stimulated when it bends away from the utricle.

d) all SCCs are stimulated.

20- All are true about utricles and saccules, EXCEPT: {د}

a) have their cells arranged in different planes.

b) are stimulated by changing the position of the head.

c) have macula as their receptors. d) respond to angular acceleration

21-Horizontal rotation of the head is likely to stimulate: {د}

a) macula of the utricle.

b) crista ampullaris of the posterior semicircular canal.

c) crista ampullaris of the anterior semicircular canal.

d) crista ampullaris of the horizontal semicircular canal.

22-When the head rotates to the right, the stimulated semicircular canal is:

a) Left horizontal canal b) Right horizontal canal

c) Left posterior canal d) Right anterior canal

23-Receptors in the semicircular canals are called:

a) Macula b) Crista ampullaris c) Organ of Corti d) Otoconia

24-The otolith organs include:

a) Anterior and posterior canals b) Utricle and saccule

c) Horizontal and vertical canals d) Ampulla and cupula

25-The receptor of the otolith organs is the:

a) Crista ampullaris b) Macula c) Cupula d) Ampulla

26-Semicircular canals or Crista ampularis detect:

a) Linear acceleration b) Static head position

c) Angular acceleration (rotation) d) Sound vibrations

27-Otolith organs or Macula detect:

a) Angular acceleration b) Linear acceleration and head position

c) Body temperature d) Blood pressure

28-Which structure detects movement in 3-dimensional planes?

a) Utricle b) Saccule c) Semicircular canals d) Cochlea

29-otolith organ is mainly:

a) Unidirectional b) Multidirectional c) Only horizontal d) Only vertical

30-semicircular canal is mainly:

a) Multidirectional b) Bidirectional c) Unidirectional d) Random

31-Maculae are located in:

a) Semicircular canals b) Utricle and saccule c) Cochlea d) Spiral ganglion

32-Depolarization of hair cells during the bending of stereocilia occurs because bending causes which of the following responses?

a. Opening channels that carry an inward K+current.

b. Closing channels that carry an outward K+ current.

c. Closing channels that carry an inward Ca2+current.

d. Closing channels that carry an inward Na+current.

e. Opening channels that carry an inward Na+current

33-Which of the following is correct concerning otolithic organ?

a. It is located only in the saccule and utricle.

b. It stimulates hair cells, transducing angular acceleration.

c. It is located only in the semicircular canals.

d. It bends stereocilia toward but not away from the kinocilium

34-The right horizontal semicircular duct ……..

a. Has crista whose kinocilia directed away from the utricle.

b. Has crista whose cupula contains statoconia.

c. Has hair cells that respond to angular acceleration.

d. Forms an angle of 30 degrees below the horizontal plane in upright position

35-Which of the following will result from rotation of the head towards one side

a. The hair cells in the canal towards which the head is rotating are depolarized

b. Same changes in hair cell memberane potential in the paired semicirciular ducts

c. Same changes in afferent firing rate of paired semicirciular ducts

d. The hair cells in the canal towards which the head is rotating are hyperpolarized

36-Semicircular canals can be stimulated by:

a) Rotation of the head b) Caloric test c) Electrical stimulation d) All of the above

37-Stimulation of a semicircular canal produces an increase in muscle tone on which side?

a) Contralateral side b) Ipsilateral side c) Both sides equally d) None

38-The macula of the utricle lies in:

a) Vertical plane b) Horizontal plane c) Oblique plane d) Sagittal plane

39-The macula of the saccule lies in:

a) Horizontal plane b) Vertical plane c) Oblique plane d) Coronal plane

40-The primary function of maculae is to detect:

a) Angular acceleration b) Linear acceleration and head position relative to gravity

c) Sound vibration d) Temperature changes

41-Utricle macula responds primarily to:

a) Vertical linear acceleration b) Horizontal linear acceleration

c) Angular acceleration d) Sound waves

42-Saccule macula responds primarily to:

a) Horizontal linear acceleration b) Vertical linear acceleration

c) Angular acceleration d) Light intensity

43-Signals from the vestibular receptors travel through which nerve?

a) Trigeminal nerve b) Glossopharyngeal nerve c) Vestibular nerve d) Facial nerve

44-Vestibular nerve fibers terminate mainly in the:

a) Cerebellar cortex b) Vestibular nuclei in the medulla

c) Red nucleus d) Hippocampus

45-The semicircular canals include the following EXCEPT:

a) Anterior canal b) Posterior canal c) Horizontal canal d) Medial canal

46-The orientation of the maculae in the otolith organs is:

a) Utricle macula is vertical; saccule macula is horizontal

b) Both utricle and saccule maculae are horizontal

c) Utricle macula is horizontal; saccule macula is vertical

d) Both utricle and saccule maculae are vertical

47-The membranous labyrinth is filled with:

a) Perilymph b) Endolymph c) Cerebrospinal fluid d) Blood

48-The space between the bony labyrinth and membranous labyrinth contains:

a) Endolymph b) Perilymph c) CSF d) Air

49-Which of the following statements is TRUE?

a) The membranous labyrinth is surrounded by endolymph

b) The membranous labyrinth is filled with perilymph

c) The membranous labyrinth is filled with endolymph and surrounded by perilymph

d) The bony labyrinth is filled with endolymph

50-Endolymph surrounds which part of a hair cell?

a) Basal part b) Apical part c) Entire cell d) None

51-Perilymph surrounds which part of a hair cell?

a) Apical part b) Basal part c) Cilia only d) None

52-Endolymph is characterized by a high concentration of:

a) Sodium b) Potassium c) Calcium d) Chloride

53-Perilymph is characterized by a high concentration of:

a) Sodium b) Potassium c) Calcium d) Chloride

54-The resting membrane potential of a hair cell is approximately:

a) 0 mV b) –30 mV c) –60 mV d) –90 mV

55-Which cilia type is single and large?

a) Stereocilia b) Kinocilium c) Microvilli d) Flagellum

56-Which cilia type is short and numerous?

a) Stereocilia b) Kinocilium c) Microvilli d) Flagellum

57-Which ion directly enters the hair cell to cause depolarization?

a) Na^+ b) K^+ c) Cl^- d) Mg^{2+}

58-Deflection of stereocilia toward the kinocilium causes:

a) Hyperpolarization b) Depolarization c) No change

59-Deflection of stereocilia away from kinocilium leads to:

a) Hyperpolarization b) Depolarization c) No change

60-What happens when stereocilia bend away from the kinocilium?

a) K^+ influx → depolarization b) K^+ channels close → hyperpolarization

c) Na^+ influx → depolarization d) Cell immediately dies

61-The depolarization in hair cells is mainly due to:

a) Na^+ influx b) K^+ influx from endolymph c) Cl^- efflux d) Ca^{2+} influx only

62-The K^+ channels in hair cells are:

a) Voltage-gated b) Ligand-gated c) Mechanically-gated d) Gap junctions

63-Functionally, hair cells in the utricle detect:

a) Vertical linear acceleration b) Horizontal linear acceleration

c) Angular acceleration d) Sound frequency

64-During rightward head rotation, the left horizontal canal is:

a) Stimulated b) Inhibited c) Not affected d) Reversed

65-When the head rotates to the left, which semicircular canal is stimulated?

a) Right horizontal canal b) Left horizontal canal

c) Left saccule d) Right posterior canal

66-Stimulation of a semicircular canal causes muscle tone to:

a) Increase on both sides b) Increase on ipsilateral side

c) Increase on contralateral side d) Decrease on ipsilateral side

67-Stimulation of a semicircular canal results in decreased muscle tone on which side?

a) Ipsilateral b) Contralateral c) Both sides d) Neither side

68-When SCCs are stimulated, the side opposite the stimulation shows:

a) Increased muscle tone b) Decreased muscle tone

c) No change d) Random fluctuation

1. C	2. B	3. B	4. A	5. D
6. C	7. B	8. B	9. A	10.C
11.A	12.A	13.A	14.D	15.A
16.B	17.A	18.C	19.D	20.D
21.D	22.B	23.B	24.B	25.B
26.C	27.B	28.C	29.B	30.C
31.B	32.A	33.A	34.C	35.A
36.D	37.B	38.B	39.B	40.B
41.B	42.B	43.C	44.B	45.D
46.C	47.B	48.B	49.C	50.B
51.B	52.B	53.A	54.C	55.B
56.A	57.B	58.B	59.A	60.B
61.B	62.C	63.B	64.B	65.B
66.B	67.B	68.B	69.	70.

Physiology of Smell & Taste

1-Which cell type in taste buds acts as a receptor for taste stimuli? {Formative2026}

a) Gustatory cells. b) Basal cells. c) Keratinocytes d) Supporting cells.

2-Which nerve transmits taste from the anterior two-thirds of the tongue? {Formative2025}

a) Vagus nerve (CN X) b) Glossopharyngeal nerve (CN IX)

c) Facial nerve (CN VII) d) Hypoglossal nerve (CN XII)

3- Which of the following is stimulate umami receptors {Final2025}

a. Sodium glutamate

4-Sour taste caused by..... {Final2023}

a) opening of k channels after binding of a substance to gustatory receptor

b) closure of k channels after binding of hydrogen

c) opening of Na channels

5-Sweet taste caused by.....

a) opening of k channels after binding of a substance to gustatory receptor

b) closure of k channels after binding of hydrogen

c) opening of Na channels

d) increase IP3 increasing intracellular Ca++

6-Bitter taste caused by.....

a) opening of k channels after binding of a substance to gustatory receptor

b) closure of k channels after binding of hydrogen

c) opening of Na channels

d) increase IP3 increasing intracellular Ca++

7-Salty taste caused by.....

a) opening of k+ channels after binding of a substance to gustatory receptor

b) closure of k+ channels after binding of hydrogen

c) opening of Na+ channels

d) increase IP3 increasing intracellular Ca++

8-Olfaction is characterized by {Final2023}

a) central adaptation b) adaptation occur at receptor level

9-Sensation of the sour taste occurs mainly by: {Formative2025}

a) opening of potassium channels in response to binding of the tasted substance to the gustatory

b) closure of sodium channels in response to binding of the tasted substance to the gustatory receptors

c) opening of sodium channels in response to binding of the tasted substance to the gustatory receptors

d) closing of potassium channels after binding of hydrogen ions to the gustatory receptors

10- Loss of smell sensation is {Final2024}

a- Anosmia b- Insomnia c- Amnesia

11- A patient is unable to taste a piece of sugar placed on the anterior part of the tongue. Which cranial nerve is likely to have a lesion : {Mid2021}

A- Hypoglossal B- Vagus C- Glossopharyngeal D- Facial

12- Taste sensation from the root of the tongue is carried by {Final2021}

A- CN VII B- CN V C- CN X D- CN IX

13- Which of the following statements about the olfactory system is TRUE? {Final2021}{Final2022}{Formative2024}

A- The receptor cells are neurons

B- The receptor cells are sloughed off and are not replaced

C- Axons of cranial nerve (CN) | are A-delta fibers

D- Axons from receptor cells synapse in the prepiriform cortex

14- A lesion of the chorda tympani nerve would most likely result in {Final2021}{Final2022}{Formative2024}

A- impaired olfactory function B- impaired vestibular function

C- impaired auditory function D- impaired taste function

15-Which of the following statements about the olfactory system is true ?. {Formative2025}

a) Axons from receptor cells synapse in the prepiriform cortex.

b) The receptor cells are sloughed off and are not replaced

c) adaptation to the olfactory stimuli is mainly central.

d) adaptation to the olfactory stimuli is mainly at the receptor cell.

16-Which of the following statements about the olfactory system is true? {د}

A) The receptor cells are neurons

B) The receptor cells are sloughed off and are not replaced

C) Axons of cranial nerve (CN) I are A-delta fibers

D) Axons from receptor cells synapse in the prepiriform cortex

E) Fractures of the cribriform plate can cause inability to detect ammonia

17-A lesion of the chorda tympani nerve would most likely result in {د}

A) impaired olfactory function B) impaired vestibular function

C) impaired auditory function D) impaired taste function E) nerve deafness

18- Sour taste is elicited by : {د}

a) alkaloids b) amino acids c) hydrogen ions d) aldehydes

1. A	2. C	3. A	4. B	5. C
6. D	7. C	8. A	9. D	10.A
11.D	12.D	13.A	14.D	15.C
16.A	17.D	18.C	19.	20.

Microbiology

Rabies

1-……is the pathognomonic clinical feature of rabies in humans. {Summar2025}

a. Aerophobia b. Hydrophobia c. Vomiting d. Paralysis

2- Post exposure prophylaxis of rabies …….. {Final2025}

a. Vaccine only b. Immunoglobulins only c. Both

3- Which of the following is false regarding rabies virus: {Formative2022}

a. The majority of cases world-wide result from bat bites

b.Infection may be prevented by active and passive immunization

c.Human Rabies vaccine is inactive vaccine

d.The animal reservoir differ from country to country

4-Which of the followings represents non-bite mode of transmission for rabies virus: {Formative2025}{قسم}

a) Rodents and rabbits b) Wild animals such as raccoons, dogs, cat.

c) Transplants of corneas taken from patients who died from cancer. d) Bat caves

5- All of the following are true about rabies virus except {Final2024}

A- Multiple genetic subtypes B- member of rhabdovirus

C- single stranded RNA D- has bullet shaped capsid

6- As regarding the pathogenesis of rabies, which of the following is TRUE? {Final2021}

A- Viraemia typically occurs in this disease

B- The clinical disease typically has flu-like manifestations only

C- The disease is transmitted only by animal bite so none bite transmission not occur

D- Infected neurons contain eosinophilic cytoplasmic inclusions called Negri bodies

E- None of the above

7-A postmortem brain biopsy of a patient exposed to a rabied demonsterated Negri bodies. Which of the following is the most likely diagnosis

a. Infectious mononucleosis b. Congenital rubella c. Mumps d. Rabies

8-Rabies is a: {ثناء}

A. Bacterial infection B. Viral infection C. Fungal infection D. Parasitic infection

9-The organism is transmitted from the bite of a rabid animal to the central nervous system via: {ثناء}

A. Arteries B. Veins C. Peripheral nerves D. Direct implantation E. Lymphatics

10-Negri bodies are: {ثناء}

A. Inflamed granulation tissue B. Aggregates of organizing fibrin

C. Epithelioid granulomas D. Foreign body granulomas

E. Intracytoplasmic inclusions

11-The cause of death in rabies is: {ثناء}

A. Muscle paralysis B. Heart failure C. Paralysis of respiratory muscles

D. Respiratory centre failure E. Septicaemia

12- Regarding rabies virus, this of the following is false regarding its replicative cycle: {قسم}

a. The virion RNA polymerase synthesizes five mRNAs that code for viral proteins.

b. Rabies virus attaches to the CD8 receptor on the cell surface.

c. progeny RNA is assembled with virion proteins to form the nucleocapsid

d. the envelope is acquired as the virion buds through the cell membrane.

13- Which of the following animals don't transmit rabies: {قسم}

a.Wild animals such as raccoons, b. Dogs and cats c.Bats. d. Rodents and rabbits.

14- Which clinical sign is considered characteristic of rabies?

a) Hydrophobia b) Jaundice c) Skin rash d) Severe diarrhea

15- Which of the following factors influences the incubation period of rabies?

a) Severity and site of the wound b) Type of rabid animal

c) Use of antibiotics d) Gender of the affected person

16- Rabies primarily affects which body system?

a) Digestive system b) Renal system

c) Central nervous system d) Circulatory system

17-Rabies virus belongs to which family?

A. Herpesviridae B. Orthomyxoviridae C. Rhabdoviridae D. Paramyxoviridae

18-Rabies is most commonly transmitted by:

A. Insect bite B. Airborne droplets

C. Fecal contamination D. Bite of a rabid animal

19-Human rabies without a history of bite may occur due to:

A. Rodent exposure B. Contaminated water

C. Inhalation of bat aerosols D. Mosquito bite

20-Rabies virus genome is:

A. Double-stranded DNA B. Double-stranded RNA

C. Single-stranded RNA D. Single-stranded DNA

21-Rabies virion contains which enzyme?

A. RNA-dependent RNA polymerase B. DNA-dependent RNA polymerase

C. Reverse transcriptase D. DNA polymerase

22-The antigenicity of rabies virus mainly resides in:

A. Nucleoprotein B. Matrix protein C. Capsid protein

D. Envelope glycoprotein spikes

23-All the following animals can transmit rabies EXCEPT:

A. Dog B. Bat C. Jackal D. Rabbit

24-Non-bite rabies exposure has been reported through:

A. Blood transfusion B. Aerosols of bat secretions

C. Contaminated food D. Sexual contact

25-Rabies has been transmitted through:

A. Kidney transplantation B. Bone marrow transplantation

C. Corneal transplantation D. Skin grafting

26-After CNS multiplication, rabies virus spreads to salivary glands through:

A. Bloodstream B. Lymphatics C. Peripheral nerves D. CSF

27-After a rabid animal bite, rabies virus first multiplies in:

A. Blood B. at the bite site C. Liver D. Spleen

28-Rabies virus reaches the central nervous system mainly by:

A. Hematogenous spread B. Lymphatic spread

C. Axonal transport along sensory nerves D. CSF circulation

29-Which statement about rabies viremia is correct?

A. Present early B. Present late C. Intermittent D. Absent

30-The usual incubation period of rabies is:

A. 1–7 days B. 1–2 weeks C. 2–16 weeks D. 6–12 months

31-Negri bodies are:

A. Nuclear inclusions B. Basophilic inclusions

C. Eosinophilic cytoplasmic inclusions D. Mitochondrial inclusions

32-Risk of developing rabies after an animal bite depends on all of the following EXCEPT:

A. Severity of exposure B. Location of the bite

C. Type of biting animal D. Age of the patient

33-The rabies vaccine contains:

A. Live attenuated virus B. Subunit vaccine

C. Recombinant protein D. Inactivated (killed) rabies virus

34-Pre-exposure rabies vaccination is recommended for:

A. All children B. Pregnant women

C. Veterinarians and zookeepers D. Patients with diabetes

35-Post-exposure rabies prophylaxis includes all EXCEPT:

A. Immediate wound cleaning B. Rabies vaccine

C. Human rabies immunoglobulin D. Live attenuated rabies virus

1. B	2. C	3. A	4. D	5. A
6. D	7. D	8. B	9. C	10.E
11.D	12.B	13.D	14.A	15.A
16.C	17.C	18.D	19.C	20.C
21.A	22.D	23.D	24.B	25.C
26.C	27.B	28.C	29.D	30.C
31.C	32.D	33.D	34.C	35.D

poliovirus

1- The poliovirus infection spread through: {Summar2025}

a) breastfeeding b) fecal-oral route c) placental route d) sexual route

2- Poliovirus multiply in ……. {Final2025}

a. Intestine b. Nerve cell c. Nasopharynx

3-Regarding poliomyelitis virus: {Formative2022}

a. The virus reaches the brain through blood

b. Asymptomatic infections lead to protective antibody production

c. The enveloped virus particles are released from infected cells by lysis

d. Herd immunity can result from vaccination with salk vaccine

4-Poliomyelitis may cause the following clinical findings Except: {Formative2024}{قسم}

a) Most polivirus infections are subclinical b) Inapparent infection.

c) causes rigid paralysis d) Aseptic meningitis.

5-Which is TRUE about Poliomyelitis {Final2023}

a) Multiply in oropharynx and intestinal tract b) Has DNA genome

6- Which of these is TRUE for the Sabin polio vaccine? {Final2021}

A- Requires four injections

B- Can be administered safely to immune- compromised patients

C- Affords secondary protection by spread to others

D- It is an inactivated vaccine

E- Cannot induce intestinal IgA

7-Mode of infection in poliomyelitis is: {ثناء}

A. Ingestion of contaminated food or drink

B. Droplet infection

C. Blood spread from a septic focus

D. Direct spread from air sinuses

8-Poliomyelitis is a: {ثناء}

A. Bacterial infection B. Viral infection C. Fungal infection D. Parasitic infection

9-Poliomyelitis is usually an infection of: {ثناء}

A. Cerebral tissue B. Cerebellar tissue C. Anterior horn cells of the spinal cord

D. Posterior horn cells of the spinal cord

E. Peripheral nerves

10-Muscle paralysis in case of poliomyelitis is due to: {ثناء}

A. Destruction of the muscle fibres by the toxins of the organisms

B. Inflammation of the muscle (myositis)

C. Inflammation of the nerves supplying the affected muscles (peripheral neuropathy)

D. Degeneration of their anterior horn cells in the spinal cord

11-Polio virus is characterized by: {قسم}

a. It is icosahedral enveloped virus.

b. The genome is double stranded DNA virus.

C. There are 3 antigenic types.

d. The virus infects animals and human.

12-The killed polio vaccine is known as:

A. Sabin vaccine B. OPV C. Salk vaccine D. None

13-The live attenuated polio vaccine is called:

A. Sabin vaccine B. IPV C. Salk vaccine D. None

14-Poliovirus causes:

A. Encephalitis B. Meningitis C. Poliomyelitis D. Guillain-Barré syndrome

15-A characteristic clinical feature of poliomyelitis is:

A. Loss of sensation B. Flaccid paralysis C. Hypertonia D. Rigidity

16-The type of paralysis seen in poliomyelitis is:

A. Spastic paralysis B. Flaccid paralysis C. Rigidity D. Parkinsonism

17-The genome of poliovirus is:

A. Double-stranded DNA B. Double-stranded RNA

C. Single-stranded RNA D. Single-stranded DNA

18-Most poliovirus infections are:

A. Fatal B. Symptomatic C. Paralytic D. Subclinical

19-Initial multiplication of poliovirus occurs in the:

A. Liver and spleen B. Lung alveoli

C. Oropharynx and Peyer's patches D. Brain

20-Poliovirus is best described as:

A. Enveloped DNA virus B. Enveloped RNA virus

C. Naked RNA virus with icosahedral symmetry D. Helical DNA virus

21-The capsid symmetry of poliovirus is:

A. Helical B. Complex C. Icosahedral D. Bullet-shaped

22-How many antigenic types of poliovirus exist?

A. One B. Two C. Three D. Four

23-Poliovirus infects naturally:

A. All mammals B. Rodents C. Birds D. Only primates

24-Poliovirus RNA is considered infectious because it can:

A. Enter nucleus directly B. Bypass the cell membrane receptor

C. Form DNA intermediate D. Integrate into host genome

25-Viremia in poliomyelitis is associated with:

A. Severe paralysis B. Loss of sensation

C. Mild fever, malaise, headache, nausea, vomiting D. Convulsions

26-The best specimen for isolation of poliovirus is:

A. Blood B. CSF C. Stool D. Urine

27-Poliovirus infection is usually acquired by:

A. Respiratory droplets B. Insect bite

C. Ingestion of contaminated food or water D. Blood transfusion

28-Poliovirus is transmitted mainly by the:

A. Fecal–oral route B. Vector bite C. Respiratory route D. Blood transfusion

29-Death in bulbar poliomyelitis usually occurs due to:

A. Cardiac arrest B. Renal failure C. Respiratory paralysis D. Septic shock

30-Primary sites of poliovirus replication are the:

A. Liver and spleen B. Lungs and bronchi

C. Oropharynx and intestinal tract D. Brain and spinal cord

31-Poliovirus is excreted mainly in the:

A. Stools B. Saliva C. Urine D. Sweat

32-Infection that stops after intestinal replication without symptoms is called:

A. Abortive infection B. Paralytic infection C. Inapparent infection D. Bulbar infection

33-Infection that stops at the viremic stage is termed:

A. Inapparent infection B. Abortive infection C. Paralytic poliomyelitis D. Bulbar poliomyelitis

34-In the CNS, poliovirus primarily destroys:

A. Posterior horn cells B. Corticospinal tract

C. Anterior horn cells of spinal cord D. Sensory ganglia

35-Bulbar poliomyelitis results from involvement of:

A. Cerebellum B. Brainstem C. Cortex D. Basal ganglia

36-The currently approved routine polio vaccination schedule consists of:

A. Two doses of OPV B. Three doses of IPV

C. Four doses of inactivated vaccine D. One lifetime dose

37-Which vaccine is contraindicated in immunocompromised individuals?

A. IPV B. Salk vaccine C. Inactivated vaccine D. Live Sabin vaccine

38-The correct ages for IPV administration are:

A. 0 , 2 weeks, 4 weeks , 6 weeks B. 6 weeks, 10 weeks, 14 weeks

C. 1 year, 5 years D. 2 months, 4 months, 6–18 months, 4–6 years

39-Which polio vaccine interrupts transmission of the virus?

A. Killed (Salk) B. Inactivated IPV C. Live attenuated (Sabin) D. Enhanced IPV

40-Induction of intestinal IgA immunity is a feature of:

A. Killed polio vaccine B. Salk vaccine C. eIPV D. Live Sabin vaccine

41-The route of administration of Salk vaccine is:

A. Oral B. Nasal C. Intradermal D. Injection

42-The route of administration of Sabin vaccine is:

A. Oral B. Nasal C. Intradermal D. Injection

43-Duration of immunity is longer with:

A. IPV B. Salk vaccine C. Killed vaccine D. Live Sabin vaccine

44-Coinfection with other enteroviruses may impair immunization with:

A. IPV B. Salk vaccine C. Inactivated vaccine D. Live Sabin vaccine

1. B	2. A	3. A	4. C	5. A
6. C	7. A	8. B	9. C	10.D
11.C	12.C	13.A	14.C	15.B
16.B	17.C	18.D	19.C	20.C
21.C	22.C	23.D	24.B	25.C
26.C	27.C	28.A	29.C	30.C
31.A	32.C	33.B	34.C	35.B
36.C	37.D	38.D	39.C	40.D
41.D	42.A	43.D	44.D	45.

Infection of the eye & ear

1- Most common causative organism of keratitis in patients using contact lenses? {Final2024}

A- Pseudomonas arginosa B- Streptococcus Aureus C- H. influenza

2-The following organism is not a common cause of eyelid infections: {Formative2025}

a) Haemophilus influenza. b) Candida albicans.

c) Pneumococci. d) Staphylococcus aureus.

3-The most common cause of viral conjunctivitis is: {Formative2024}

a) Respiratory syncytial virus b) Adenovirus c) Poliovirus d) Corona virus

4- Keratitis mean microbial infection of {Formative2022}

a. cornea b. retina. c. eye lid d. conjunctiva

5- As regarding ocular trachoma; choose the wrong answer ? {Final2021}

A- This disease is endemic in Egypt

B- The disease is characterized by acute conjunctivitis only and the cornea is not involved

C- It is caused by Chlamydia trachomatis biovar TRIC serotypes A, B and C

D- Infection is transmitted from eye to eye by fingers, flies

E- None of the above

6-Which of the following is NOT TRUE about trachoma? {Final2023}

a) Caused by subtypes d- k b) Can cause blindness

c) Produce mucopurulent conjunctivitis

7-Which of the following is the most common type of eye infection?

A) Keratitis B) Conjunctivitis C) Uveitis D) Endophthalmitis

8-Infection of the eyelid margin is called:

A. Conjunctivitis B. Keratitis C. Blepharitis D. Endophthalmitis

9-Infection of the conjunctiva is called:

A. Blepharitis B. Keratitis C. Endophthalmitis D. Conjunctivitis

10-Infection of the cornea is termed:

A. Conjunctivitis B. Blepharitis C. Endophthalmitis D. Keratitis

11-The most common bacterial cause of keratitis in contact lens users is:

A) Staphylococcus aureus B) Pseudomonas aeruginosa

C) Neisseria gonorrhoeae D) Haemophilus influenzae

12-The most common causative organism of eyelid infections is:

A. Streptococcus pyogenes B. Haemophilus influenzae

C. Staphylococcus aureus D. Pseudomonas aeruginosa

13-The most common causes of eyelid infections include all of the following EXCEPT:

A. Neisseria gonorrhoeae B. Pneumococci

C. Haemophilus influenza D. Staphylococcus aureus

14-Which of the following is a common bacterial cause of adult conjunctivitis?

A) Neisseria gonorrhoeae B) Chlamydia trachomatis

C) Staphylococcus aureus D) Streptococcus pyogenes

15-The most common viral cause of conjunctivitis is:

A) Herpes simplex virus B) Adenovirus C) Measles D) Varicella-zoster virus

16-The most common viral cause of keratitis is:

A) Adenovirus B) Herpes simplex virus C) Enterovirus D) Varicella-zoster virus

17-The most common bacterial organisms causing endophthalmitis include:

A) Staph aureus, Staph epidermidis, Strept pneumoniae

B) Pseudomonas aeruginosa, Haemophilus influenzae, Candida

C) Adenovirus, Enterovirus, HSV

D) Aspergillus, Fusarium, Candida

18-The most common bacterial cause of acute otitis media is:

A) Streptococcus pneumoniae B) Haemophilus influenzae

C) Staphylococcus aureus D) Pseudomonas aeruginosa

19-All of the following are common bacterial causes of acute otitis media EXCEPT:

A) Streptococcus pneumonia B) Haemophilus influenzae

C) Staphylococcus aureus D) Aspergillus spp

1. A	2. B	3. B	4. A	5. B
6. A	7. B	8. C	9. D	10.D
11.B	12.C	13.A	14.C	15.B
16.B	17.A	18.A	19.D	20.

Pharmacology

ATYPICAL SEDATIVE-HYPNOTICS

1- All the following antipsychotic drugs cause extrapyramidal side effects except: {Summar2025}

A- Chlorpromazine B-Haloperidol C- Clozapine D- Phenothiazine

2- All of the following are indications of phenothiazines except …. {Final2025}

a. Psychosis b. Parkinsonism

3- Buspirone is …… {Final2025}

a. 5HT1A partial agonist used as anxiolytic

4-All the following drugs are atypical antipsychotics except: {Formative2025}

a) olanzapine b) risperidone, c) clozapine, d) chlorpromazine

5- The extrapyramidal side effects of clozapine are exerted by blocking of:- {Final2024}

a) Histamine receptors b) 5-HT2 receptors

c) Dopamine receptors in nigrostriatal system. d) Dopamine receptors in the mesolimbic system.

6-Atypical antipsychotic is... {Final2023}

a) Chlorpromazine b) Haloperidol c) Clozapine

7- Which of the following is TRUE about Buspirone ? {Final2022}

A- It is a partial agonist on serotonin receptors B- Tricyclic antidepressant

C- Local anesthetics D- None of the above

8-The antipsychotic effect of clozapine is exerted by blocking of:- {Formative2024}

a) Histamine receptors b) 5-HT2 receptors

c) Dopamine receptors in nigrostriatal system.

d) Dopamine receptors in the mesolimbic system.

9- Indicate the antipsychotic drug, which is a butyrophenone derivative ? {Final2021}{Final2022}

A- Droperidol B- Thioridazine C- Sertindole D- Fluphenazine

10- Tardive dyskinesia is the result of: {Final2021}

A- Degeneration of dopaminergic and cholinergic fibers

B- Hyperactive dopaminergic state in presence of dopamine blockers

C- Degenerations of histaminergic fibers

D- Supersensitivity of cholinergic receptors in the caudate putamen

11-Phenothiazines include all EXCEPT:

A. Chlorpromazine B. Thioridazine C. Trifluperazine D. Clozapine

12-Which of the following is a typical antipsychotic?

A. Olanzapine B. Clozapine C. Haloperidol D. Risperidone

13-Which of the following is an atypical antipsychotic?

A. Chlorpromazine B. Haloperidol C. Thioridazine D. Clozapine

14-Typical antipsychotics can cause which of the following side effects?

A. Extrapyramidal symptoms B. Tardive dyskinesia

C. Neuroleptic malignant syndrome D. All of the above

1. C	2. B	3. A	4. D	5. C
6. C	7. A	8. D	9. A	10.B
11.D	12.C	13.D	14.D	15.

Parkinsonism

1-The following drug is combined with Levodopa to increase its plasma level: {Summar2025}

a) Bromocriptine b) Ropinirole c) Carbidopa d) Pramipexole

2- The most eftective drug in Parkinsonism is {Final2024}

A- Bromocriptine B- Selegiline C- Levodopa D- Biperiden

3- Which drug abolishes therapeutic effect of Ldopa? {Final2024}

A- Chlorpromazine B- Pyridoxine C- Metclopramide

4- Which of the following is dopamine agonist ? {Final2025}

a. Levodopa b. Pramipexole

5-Pure D2 receptor agonist {Final2023}

a) Ropinirole b) Pramipexole c) Bromocriptine

6- Regarding connections of the basal ganglia, which one of these neurotransmitters Is consider with inhibition ? {Final2021}

A- Glutamate B- Dopamine 1 C- Dopamine 2 D- Acetylcholine

7-Value of combination of carbidopa with levodopa is.... {Final2023}

a) Stimulate peripheral decarboxylation b) reduce levodopa induced dyskinesia

c)enter CNS and increase dopamine d) none of the above

8- The redness, tenderness, and painful swollen feet is the side effect of the following drug {Mid2021}

A- Amantadine B- Benztropine C- Bromocriptine D- Levodopa

9-The following drug causes swollen tender bluish discolored feet in patient with Parkinsonism {Formative2024}{Summar2025}

a) Selegiline b) Bromocriptine c) Benztropine d) Amantadine

10- The most effective single drug in Parkinsonism is {Final2021}{Final2022}

A- Bromocriptine B- Selegiline C- Levodopa D- Biperiden

11-All of the following are contraindications to levodopa EXCEPT:

A. Psychotic illness B. Closed-angle glaucoma

C. Open-angle glaucoma D. Malignant melanoma

12-Levodopa should NOT be given in all of the following conditions EXCEPT:

A. Cardiac dysrhythmia B. Active peptic ulcer

C. Parkinsonism with tremor D. Psychotic illness

13-Which drug is combined with levodopa to prevent its peripheral decarboxylation?

A. Selegiline B. Entacapone C. Carbidopa D. Amantadine

14-Which drug is a direct precursor of dopamine?

A. Bromocriptine B. Levodopa C. Amantadine D. Selegiline

15-Parkinson's syndrome, tardive dyskinesia, antimuscarinic actions, orthostatic hypotension, galactorrhea are possible adverse effects of:

a. Chlorpromazine b. Olanzapine c. Risperidone d. Quetiapine

16-Which antiemetic commonly causes Parkinsonian symptoms?

A. Ondansetron B. Domperidone C. Metoclopramide D. Promethazine

17-Which of the following drugs is used in the treatment of Parkinsonian disorders

a. Haloperidol b. Ipratropium c. Selegiline d. Phenytoin

18-Carbidopa enhances the effect of levodopa mainly by:

A. Crossing the BBB and increasing CNS dopamine B. Inhibiting COMT in the brain

C. Inhibiting peripheral DOPA decarboxylase D. Stimulating dopamine receptors

19-Levodopa is usually given with carbidopa to:

A. Increase CNS dopamine breakdown B. Reduce peripheral metabolism of levodopa

C. Stimulate dopamine receptors D. Prevent COMT inhibition

20-Drug-induced Parkinsonism is mainly caused by blockade of which receptor?

A. D_1 B. D_2 C. GABA D. NMDA

21-Bromocriptine acts mainly on which dopamine receptor subtype?

A. D_1 B. D_2 C. D_3 D. D_4

22-Bromocriptine is used in all of the following EXCEPT:

A. Parkinson's disease B. Hyperprolactinemia C. Acromegaly D. Alzheimer's disease

23-Which drug is an atropine substitute used in Parkinson's disease?

A. Levodopa B. Trihexyphenidyl C. Bromocriptine D. Selegiline

24-Which antipsychotic has the highest risk of causing drug-induced Parkinsonism?

A. Clozapine B. Risperidone C. Haloperidol D. Quetiapine

25-α-Methyldopa causes Parkinsonism by:

A. Blocking dopamine receptors B. Destroying dopamine neurons

C. Inhibiting dopamine synthesis D. Increasing dopamine metabolism

26-Which of the following is an ergot-derived dopamine agonist?

A. Pramipexole B. Ropinirole C. Amantadine D. Bromocriptine

27-Combination of levodopa with carbidopa results in:

A. Increased peripheral dopamine formation B. Reduced plasma levodopa levels

C. Increased plasma levodopa levels D. Shortened levodopa half-life

28-Rasagiline is useful in Parkinson's disease because it:

A. Stimulates D_2 receptors B. Inhibits MAO-B

C. Blocks COMT D. Increases acetylcholine

29-Which drug is a COMT inhibitor?

A. Selegiline B. Rasagiline C. Entacapone D. Amantadine

30-Dopamine is ineffective in Parkinson's disease when given systemically because it:

A. Is rapidly metabolized by COMT B. Does not cross the blood–brain barrier

C. Is poorly absorbed from the gut D. Causes severe hypotension

31-Which drug is the immediate metabolic precursor of dopamine?

A. Tyrosine B. Dopamine C. Levodopa D. Norepinephrine

32-Pyridoxine reduces the therapeutic effect of levodopa by:

A. Inhibiting dopamine receptors B. Enhancing peripheral metabolism of levodopa

C. Blocking levodopa absorption D. Inhibiting COMT

33-Levodopa should NOT be given with MAO-A inhibitors because of risk of:

A. Serotonin syndrome B. Severe hypotension

C. Hypertensive crisis D. Neuroleptic malignant syndrome

34-Therapeutic uses of bromocriptine include all EXCEPT:

A. Parkinson's disease B. Acromegaly

C. Migraine prophylaxis D. Hyperprolactinemia

35-Which drug may affect valves in Parkinson's therapy?

A. Levodopa B. Bromocriptine C. Pergolide D. Pramipexole

1. C	2. C	3. B	4. B	5. C
6. C	7. D	8. A	9. D	10.C
11.C	12.C	13.C	14.B	15.A
16.C	17.C	18.C	19.B	20.B
21.B	22.D	23.B	24.C	25.C
26.D	27.C	28.B	29.C	30.B
31.C	32.B	33.C	34.C	35.C

Pharmacotherapy of epilepsy

1- All of the following are broad-spectrum anticonvulsant EXCEPT: {Summar2025}

a) Topiramate b) Lamotrigine c) Valproic acid d) Carbamazepine

2- The drug of the first choice in treatment of Grand Mal Epilepsy is: {Summar2025}

a) Carbamazepine b) Phenytoin c) Diazepam d) Ethosuximide

3- Which of the following is used in treatment of petit mal seizures but worsen grand mal seizures? {Final2025}

a. Valporate b. Ethosuximide

4-Which type of epilepsy can be worsened by phenytoin?

A. Partial seizures B. Grand mal seizures

C. Petit-mal (absence) seizures D. Tonic seizures

5-It is the drug of choice in the treatment of petit-mal epilepsy (absence seizure): {Merit}

A. Topiramate B. Lamotrigine C. Ethosuximide D. Phenobarbital

6-First-line treatment of status epilepticus is:

A. Phenytoin B. Diazepam C. Carbamazepine D. Valproate

7-All of the following are used in status epilepticus Except

A. Phenytoin B. Diazepam C. Clonazepam D. Valproate

8-A 7-year-old girl has been having 5-10 second staring spells several times a day while in class. During these spells, the girl exhibits small amplitude hand motions, and yet becomes fully alert immediately afterwards What would be a drug of first choice for treating this patient's seizure disorder? {Final2022}{Final2024}{Formative2024}

a) Diazepam b) Carbamazepine c) phenytoin d) Ethosuximide

9-All are true about Phenytoin except …… {Final2025}

a. it's used in treatment of ventricular arrhythmia b. Its elimination is dose dependent

c. it's used in treatment of Bibolar disorder

10- Which drug used in mixed petit and grand mal epilepsy? {Final2024}

A- valproate

11-Therapeutic uses of phenytoin include all EXCEPT:

A. Status epilepticus B. Partial seizures C. Petit-mal seizures D. Ventricular arrhythmias

12-Which of the following drugs would be the most appropriate choice for treating generalized tonic-clonic seizure, due to its ability to block Na channels in a frequency-dependent manner? {Formative2024}

a) Carbamazepine b) phenobarbital c) Ethosuximide d) Diazepam

13- Main side effects of valproate {Final2023}

a. spina bifida pancreatitis

14- A patient has taken carbamazepine & gabapentin but seizures continued, when we add phenytoin, what happens to its plasma level compared to when use as monotherapy ? {Final2023}

a) lower b) higher c) no difference

15- One of the following is among the therapeutic uses of benzodiazepine {Formative2022}

A. Clonazepam is useful in absence seizures and diazepam used in status epilepticus

B. Alprazolam has been used in the treatment of panic disorders

C. Midazolam, diazepam and lorazepam are the most frequently used benzodiazepine in the perioperative period

D. All of the above

16- Indicate an anti-absence drug {Final2021}{Final2022}

A- Valproate B- Phenobarbital C- Carbamazepine D- Phenytoin

17- Indicate an antiseizure drug, inhibiting central effects of excitatory amino acids ? {Final2021}{Final2022}

A- Ethosuximide B- Lamotrigine C- Diazepam D- Tiagabine

18- One of these drugs is new for epilepsy ? {Final2022}

A- Phenytoin B- Valproate C- Carbamazine D- Lamotrigine

19- Diazepam can be used in treatment of the following disorders EXCEPT {Final2021}

A- Status epilepticus B- Absence seizures C- Anxiety disorders D- Sleep disorders

20-A 53-year-old woman with seizure disorder, bipolar disorder and trigeminal neuralagia presents to her primary care physician for follow-up and treatment. She has no new complaints. Which of the following medications may serve to treat all of her earlier mentioned problems?

a. Lacosamide b. Carbamazepine c. Gabapentin. d.. Ethosuximide

21-A 78-year-old man with long history of seizure disorder controlled with phenytoin presents to his dentist for routine follow-up. Which of the following findings must the dentist be concerned about and evaluate for?

a. Gingival overgrowth b. Exposed nerve roots c. Teeth erosion. d. Jaw bone exposure

22-The following antiepileptic drug is useful for trigeminal neuralgia: {Merit}

A. Phenytoin B. Valproic acid C. Gabapentin D. Carbamazepine

23-Which of the following is a common long-term adverse effect of phenytoin in children?

A. Gingival hyperplasia B. Valvular heart disease C. Cataract formation D. Alopecia

24-Phenytoin's antiarrhythmic effect is mainly for:

A. Supraventricular tachycardia B. Ventricular arrhythmias, especially digitalis-induced

C. Bradycardia D. Atrial fibrillation

25-Phenytoin is used for all of the following EXCEPT:

A. Partial and grand mal seizures B. Status epilepticus

C. Trigeminal neuralgia (if carbamazepine fails) D. Petit-mal (absence) seizures

26-Carbamazepine is avoided in which type of epilepsy?

A. Partial seizures B. Grand mal seizures

C. Petit-mal (absence) seizures D. All of above

27-Valproate is NOT suitable for:

A. Mixed petit-mal + grand mal epilepsy B. Status epilepticus

C. Absence seizures D. Partial seizures

28-Carbamazepine is a first-line drug for:

A. Absence seizures B. Partial and generalized tonic-clonic seizures

C. Myoclonic seizures D. Febrile seizures

29-Phenytoin is indicated for all of the following EXCEPT:

A. Ventricular arrhythmias (especially digitalis-induced) B. Tonic-clonic seizures

C. Migraine prophylaxis D. Trigeminal neuralgia

30-All of the following are long-term adverse effects of phenytoin EXCEPT:

A. Gingival hyperplasia B. Cleft palate C. Hirsutism D. Valvular heart disease

31-The primary mechanism of carbamazepine's antiepileptic effect is:

A. GABA receptor agonism B. Blockade of sodium channels in neuronal cell membranes

C. Inhibition of calcium channels D. NMDA receptor antagonism

32-Carbamazepine is also used in psychiatry for:

A. Schizophrenia B. Bipolar disorder (manic-depressive illness)

C. Anxiety disorders D. Depression only

33-Carbamazepine is used for all of the following EXCEPT:

A. Partial seizures B. Generalized tonic-clonic seizures

C. Trigeminal neuralgia D. Petit-mal (absence) seizures

34-Valproate is the drug of choice for:

A. Grand mal only B. Petit-mal only

C. Mixed petit-mal + grand mal epilepsy D. Myoclonic seizures only

35-Valproate is effective in all types of epilepsy EXCEPT:

A. Absence seizures B. Partial seizures C. Tonic-clonic seizures D. Status epilepticus

36-Valproate is teratogenic, causing:

A. Cleft palate B. Spina bifida C. Cardiac defects only D. Microcephaly

37-All of the following are known adverse effects of valproate EXCEPT:

A. Fulminant hepatitis B. Sedation and ataxia C. Cleft palate D. Tremors

1. D	2. A	3. B	4. C	5. C
6. B	7. D	8. D	9. C	10.A
11.C	12.A	13.A	14.A	15.D
16.A	17.B	18.D	19.B	20.B
21.A	22.D	23.A	24.B	25.D
26.C	27.B	28.B	29.C	30.D
31.B	32.B	33.D	34.C	35.D
36.B	37.C	38.	39.	40.

SEDATIVES-HYPNOTIC DRUGS

1- Life saving drug in benzodiazepine overdose ? {Final2024}

A- flumazenil B- Zolipedine

2- ALL of the following are of benzodiazepine receptors agonists EXCEPT: {Formative2022}

A. Zolpidem B. Buspirone C. Zaleplon D. Benzodiazepines.

3-One of the following is true regarding buspirone: {Formative2024}

a) It is effective in severe form of anxiety b) It is partial agonist to 5HT1A receptor

c) It facilitates the action of the GABA d) It has an anticonvulsant and muscle relaxant effects

4- One of the following drugs acts as inverse agonists at benzodiazepine receptor: {Final2023}{Summar2025}

a-Clonazepam b- Zolpidem c-Flumazenil d- B-carbolines

5-One of the following is true regarding mechanism of action of benzodiazepine {Formative2024}

a) It antagonizes the inhibitory effect mediated by the GABA

b) It increases the duration of chloride channel-opening

c) It has GABAmimetic effect

d) It increases in the frequency of chloride channel-opening

6-Benzodiazepines have one of the following pharmacological advantages compared to barbiturates:- {Formative2025}

a) More tendency for abuse and dependence. b) More depressant effect on respiration.

c) It have better therapeutic index d) It is an inducers of the liver microsomal enzyme.

7- Which of the following Z-compounds can be used safely for treatment of sleep-onset insomnia ? {Final2021}{Final2022}

A- Ramelteon B- Buspirone C- Diazepam D- Zolpidem

8-Indicate the hypnotic benzodiazepine, which has the shortest elimination half-life:

a. Temazepam. b. Diazepam. c. Triazolam. d. Flurazepam

9-Indicate the competitive antagonist of BZ receptors:

a. Temazepam. b. Zolpidem. c. Flumazenil. d. Flurazepam.

10-Select a hypnotic drug, which is a benzodiazepine derivative:

a. Secobarbital. b. Flurazepam c. Phenobarbitone d. Chloral hydrate

11-Which of the following is a benzodiazepine receptor antagonist?

A) Zaleplon B) Lorazepam C) β-carbolines D) Flumazenil

12-The effect of benzodiazepine inverse agonists is to:

A) Enhance GABA-mediated inhibition B) Produce anxiety and seizures

C) Block barbiturate action D) Act as sedatives

13-Compared to barbiturates, benzodiazepines are preferred because they:

A) Have a higher risk of respiratory depression B) Are safer in overdose

C) Have no sedative effect D) Are less effective as hypnotics

14-The primary mechanism of action of benzodiazepines is:

A) Direct activation of GABA-A receptors without GABA

B) Potentiation of GABA-mediated inhibitory effects

C) Inhibition of dopamine receptors D) Blockade of NMDA receptors

15-Compared to barbiturates, benzodiazepines have all of the following pharmacological advantages EXCEPT:

A) Less depressant effect on respiration B) Less tendency for abuse and dependence

C) Not liver enzyme inducers, thus fewer drug interactions D) Greater risk of fatal overdose

16-Which of the following is TRUE regarding benzodiazepines compared to barbiturates?

A) Higher tendency for dependence B) Induce liver enzymes

C) Safer in overdose D) Narrow therapeutic index

17-Which benzodiazepine is commonly used for skeletal muscle relaxation in patients with cerebral palsy?

A) Diazepam B) Clonazepam C) Lorazepam D) Zolpidem

18-Which of the following correctly matches the interaction type with its effect?

A) Agonist → anxiety induction B) Antagonist → potentiates GABA

C) Inverse agonist → produces anxiety and seizures D) Antagonist → increases sedation

19-Which of the following are agonists of benzodiazepine receptors?

A) Zolpidem and Zaleplon B) Melatonin receptor agonists

C) Orexin antagonists D) Barbiturates

20-Which of the following is classified as an atypical sedative-hypnotic?

A) Zaleplon B) Melatonin receptor agonist C) Barbiturates D) Chloral hydrate

21-The main clinical use of benzodiazepine receptor agonists is:

A) Inducing seizures B) Treating anxiety

C) Blocking benzodiazepine action D) Enhancing barbiturate toxicity

22-Which benzodiazepine is most commonly used for preanesthetic medication?

A) Midazolam B) Zaleplon C) Diazepam only D) Triazolam

23-Flumazenil is primarily used for:

A) Treating barbiturate overdose B) Treating benzodiazepine overdose

C) Inducing sedation D) Producing seizures

24-Which of the following is a short-acting benzodiazepine?

A) Diazepam B) Temazepam C) Triazolam D) Flurazepam

25-Which of the following is a benzodiazepine inverse agonist?

A) β-carbolines B) Zaleplon C) Diazepam D) Flumazenil

26-Alprazolam is especially indicated for:

A) Generalized anxiety disorder B) Panic disorders C) Insomnia D) Muscle relaxation

27-The skeletal muscle relaxant effect of clonazepam is useful in:

A) Epilepsy B) Cerebral palsy C) Insomnia D) Anxiety

28-Which of the following benzodiazepines have anticonvulsant properties?

A) Clonazepam, Nitrazepam, Nordazepam B) Zolpidem, Zaleplon

C) Ramelteon, Suvorexant D) Flumazenil

29-Oxazepam belongs to which class based on elimination half-life?

A) Short-acting B) Intermediate-acting C) Long-acting D) Ultra-long acting

30-Which of the following is an intermediate-acting benzodiazepine?

A) Flurazepam B) Temazepam C) Quazepam D) Diazepam

31-In status epilepticus, the benzodiazepine of choice is:

A) Diazepam B) Clonazepam C) Temazepam D) Zolpidem

32-Which benzodiazepine is preferred for absence seizures?

A) Diazepam B) Midazolam C) Lorazepam D) Clonazepam

33-Which of the following hypnotics are effective in relieving sleep-onset insomnia?

A) Temazepam and Flurazepam B) Zaleplon and Zolpidem

C) Diazepam and Lorazepam D) Clonazepam and Nitrazepam

34-Melatonin receptor agonists include:

A) Zolpidem and Zaleplon B) Ramelteon and Tasimelteon

C) Diazepam and Lorazepam D) Flurazepam and Temazepam

1. A	2. B	3. B	4. D	5. D
6. C	7. D	8. C	9. C	10.B
11.D	12.B	13.B	14.B	15.D
16.C	17.B	18.C	19.A	20.B
21.B	22.A	23.B	24.C	25.A
26.B	27.B	28.A	29.A	30.B
31.A	32.D	33.B	34.B	35.

Biochemistry

Brain Metabolism

1-In hypoxic states, there is increase in {Formative2024}{سؤال في المحاضرة}

a) brain amino acids b) spinal fluid lactate c) spinal fluid glucose d) brain glucose

2-The main substrate used for brain energy production is: {Formative2024}{سؤال في المحاضرة}

a) glucose b) starch c) amino acids d) mannose

3-Consciousness loss during Cerebral blood flow interruption due to: {سؤال في المحاضرة} {Formative2024}

a) Lack of serotonin b) Lack of glutamate c) Lack of glucose d) Lack of oxygen

4- Which of the following substances depletion can lead to neuronal degeneration? {Summar2025}

a-glucose b-mannose c-cholesterol d-glutamate

5-Which of the following can cross blood brain barrier easily? {Formative2025}

a) fructose b) glucose c) fatty acids d) lactose

6- Mannose to be used for the brain must be converted to {Final2025}

a. Fructose -6- phosphate b. Glucose

7- Amino acid in the brain important for detoxification of ammonia and act as a neurotransmitter. {Final2025}

a. Glutamate

8- Which of the following cross BBB easily ? {Final2024}

A- mannose B- fructose C- lacose D- pyruvate

9- When blood glucose is low, the brain uses which of the following for energy? {Final2021}

A- cholesterol B- glutamate C- glycogen D- fatty aids E- starch

10- Which of the following substrates can be used for brain energy production ? {Mid2021}

A- Glucose B- Mannose C- Amino acids D- A&B E- Starch

11-The amount of oxygen utilized by the brain out of total consumption is approximately:

a. 40%. b. 10%. c. 30%. d. 20%. e. 50%.

12-Which of the following organ has an obligatory requirement for glucose as the fuel:

a. Kidney. b. Liver. c. Brain. d. Heart.

13-Which amino acid is found in higher concentrations in the brain? {Final2023}

a) Tyrosine b) Phenyl alanine c)Glutamate d) Tryptophane

14- Gamma-Aminobutyrate (GABA) is synthesized from: {Summar2025}

a- tyrosine b-glutamate c-glycine d-serine

15- Glutamate: {Final2024}

a- Used for Ammonia detoxication b- Used For GABA synthesis and Ammonia detoxication

16- increase level of GABA in the brain increases which of the following {Final2024}

a. Cl- b. Glucose

17-Brain contains highest concentration of which of the following amino acids: {Formative2023}

a-serine b-phenyl alanine c-glutamate d-threonine

18-The major substrate used for energy production in cerebral tissue is:

A) Fatty acids B) Ketone bodies C) Glucose D) Amino acids

19-Cerebral tissue utilizes glucose mainly from:

A) Venous blood B) Cerebrospinal fluid C) Arterial blood D) Stored glycogen

20-Which sugar can cross the blood–brain barrier easily and enter glycolysis as fructose- 6-phosphate?

A) Glucose B) Galactose C) Mannose D) Fructose

21-Which of the following has limited permeability across the BBB and cannot directly serve as a cerebral energy substrate?

A) Glucose B) Lactate C) Mannose D) Oxygen

22-During hypoxia, the rate of glycolysis in cerebral tissue:

A) Decreases B) Remains unchanged C) Slightly decreases D) Increases

23-Increased lactic acid production during hypoxia results mainly in:

A) Alkalosis B) Intracellular acidosis

C) Increased ATP synthesis D) Decreased glucose utilization

24-The metabolic changes seen during hypoxia are explained by the:

A) Cori cycle B) Crabtree effect C) Pasteur effect D) Hexose monophosphate shunt

25-Loss of consciousness occurs primarily due to:

A) Decreased blood glucose B) Reduced insulin levels

C) Interruption of cerebral blood flow D) Increased lactate production

26-Which of the following conditions does NOT reduce cerebral oxygen uptake?

A)Insulin-induced hypoglycemia B)Diabetic coma C)Cerebral tumors D)Hyperthyroidism

27-During hypoxia, cerebral glucose concentration:

A) Increases B) Remains unchanged C) Decreases D) Converts to glycogen

28- Synthesis of GABA from glutamate need {Mid2023}

A- PLP B- NAD

29-The catecholamines include: {Formative2026}

a) Epinephrine, serotonin, and dopamine

b) Epinephrine, norepinephrine, dopamine, and serotonin

c) Dopamine, serotonin, and acetylcholine

d) Epinephrine, norepinephrine, and dopamine

30-Deficiency of which of the following can cause depression? {Formative2025}

a) Serotonin b) Acetylcholine c) GABA d) Catecholamines

31- Depression may be caused by {Mid2024}{Summar2025}

a) Excess of dopamine b) Excess of serotonin

c) Insufficient serotonin d) Insufficient acetylcholine

32- Serotonin is inactivated to ..... By ..... {Mid2021}

A- 5 hydroxy tryptophan, serotonin hydroxylase B- 5 hydroxy tryptophan, serotonin oxidase

C- 5 hydroxy tryptamine, serotonin hydroxylase

D- 5 hydroxy indole acetic acid, serotonin hydroxylase

E- 5 hydroxy indole acetic acid, monoamino oxidase

33- Histamine is derived from {Mid2021}

A- Tyrosine B- Histidine C- Adrenaline D- Glutamate E- Dopamine

34- Which one is the cause of wilsons disease ? {Final2021}{سؤال في المحاضرة}

A- High copper B- Low copper C- High O2 D- Low O2

35- Dopamine formed from: {Formative2023}

a) Glutamate b) Tryptophan c) Tyrosine d) Glycine

36- The 5-HT receptors related to which of the following neurotransmitters? {Formative2024}

a. Serotonin b. Dopamine c. Acetylcholine d. Tryptophan

1. B	2. A	3. D	4. A	5. B
6. A	7. A	8. A	9. C	10.D
11.D	12.C	13.C	14.B	15.B
16.A	17.C	18.C	19.C	20.C

21.B	22.D	23.B	24.C	25.C
26.D	27.C	28.A	29.D	30.A
31.C	32.E	33.B	34.A	35.C
36.A	37.	38.	39.	40.

Parasitology

1- The vector of Onchocerca volvulus is: {Summar2025}

a) Female Anopheles b) Chrysops c) Simulium (black fly) d) Glossina palpalis

2- It is associated with uveitis or ulceration of the cornea (keratitis). {Summar2025}

a) Acanthamoeba culbertsoni b) Trypanosoma cruzi

c) Toxoplasma gondii d) Naegleria fowleri

3- Retinal blind spot is caused by: {Final2024}

a- Toxoplasma gondii

4- Ocular sparganosis is caused by ingestion of {Final2024}

a. Procercoid of diphyllobthrium mansoni b. Plerocercoid of diphyllobothrium latum

c. Plerocercoid larva of diphyllobothrium mansoni d. Egg of Diphyllobothrium mansoni

5-Coenurus cerebralis: {Formative2025}

a) Affects the brain and spinal cord. b) Is the larval stage of Taenia solium

c) Infection to man is caused by autoinfection. d) Caused by Echiococcus granulosus

6-Route of infection in Hydatic cyst {Final2023}

a) Ingestion of egg of Echinococcus granulosus b) Ingestion of hydatic cyst

7-Onchocerca vector {Final2023}

a) Black fly (simulium) b) Mango fly

8-The vector of Onchocerca volvulus is: {Formative2025}

a) Chrysops b) Female Anopheles c) Glossina palpalis d) Simulium

9-Riverine blindness can be caused by: {Formative2023}

a) Onchocerca volvulus b) Loa loa c) Taenia solium d) Toxoplasma gondii

10- Romana's sign can be caused by: {Formative2023}

a) Loa loa b) Acanthamoeba spp c) Trypanosoma cruzii d) Onchocerca volvulus

11- Bachman intradermal test used for diagnosis of {Final2021}

A- Schistosomisis B- Trichinelliasis C- Hydatid disease

D- Kala azar E- Sparganosis

12- Bachman intradermal test used for diagnosis of {Final2024}

A- Schistosomisis B- Trichinella spiralis C- Hydatid disease D- Sparganosis

13- intradermal test used to diagnosis of trichinella spiralis is {Final2022}

A- bachman test B- casoni test C- mazotti test D- none of the above

14- Onchocerca volvulus causes {Final2022}

A- Cerebral hydatid cyst B- cysticercosis C- chaases D- riverblind

15- the African eye warm is {Final2021}

A- Onchocerca volvulus B- Toxoplasma gondil C- Loa loa D- Acabthamaeba

1. C	2. A	3. A	4. C	5. A
6. A	7. A	8. D	9. A	10.C
11.B	12.B	13.A	14.D	15.C